

Research paper

Mitigating dataset imbalance using image-based stable diffusion and feature-level SMOTE for solar panel classification with CNNs

Raj Kumar^a, Yong-Woon Kim^b, Yung-Cheol Byun^{c,*}^a Department of Electronic Engineering, Institute of Information Science and Technology, Jeju National University, Jeju-si 63243, South Korea^b Department of Computer Engineering, Jeju National University, Jeju 63243, South Korea^c Department of Computer Engineering, Major of Electronic Engineering, Jeju National University, Institute of Information Science & Technology, Jeju 63243, South Korea

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ABSTRACT

Imbalanced datasets pose a significant challenge in developing predictive models for solar panel fault detection, particularly for identifying dust accumulation on photovoltaic (PV) modules. This study addresses class imbalance in 842 polycrystalline silicon (poly-Si) PV module images (502 clean, 340 dusty) from the publicly available SolNET dataset collected across dusty regions of Bangladesh. All images were processed as 3-channel RGB inputs and resized to 224×224 pixels for CNN training (camera metadata and capture parameters such as irradiance, temperature, and distance are not provided in the dataset release). Two complementary augmentation strategies were employed: feature-level SMOTE (Synthetic Minority Over-Sampling Technique) and image-level Stable Diffusion augmentation. Stable Diffusion generated 162 photorealistic dusty-panel images, retained through a two-stage validation protocol combining visual screening and feature-space similarity testing (FID = 63.86, KID = 0.0220, pHash duplicates = 0 %), ensuring realism and diversity. Three convolutional neural networks (VGG-16, ResNet50, and MobileNetV3) were evaluated using stratified 10-fold cross-validation, with performance assessed by accuracy, F1-score, Cohen's κ , and Matthews correlation coefficient (MCC). Results showed consistent improvements; for example, VGG-16 accuracy increased from 84.6 % (imbalanced) to 93.5 % (SMOTE) and 97.6 % (Stable Diffusion), while ResNet50 achieved 98.9 % with Stable Diffusion. Cross-validation confirmed robust performance (VGG-16: 95.6 % $\pm$ 1.7 % accuracy, κ = 0.91 $\pm$ 0.02, MCC = 0.93 $\pm$ 0.02), and Wilcoxon signed-rank testing verified statistical significance ($p < 0.05$). Constraints include reliance on a single public dataset, binary dust/clean labeling, uniform resizing, and restriction to poly-Si modules, which may limit generalization to other PV technologies and field conditions. Overall, the findings indicate that combining feature-level oversampling with diffusion-based image augmentation reduces false negatives and improves sensitivity to dusty panels, motivating further validation on larger, multi-technology PV datasets.

1. Introduction

Driven by concerns about climate change and global warming, the global renewable energy capacity reached approximately 4448 GW by the end of 2024, of which photovoltaic (PV) solar energy accounts for approximately 1749 GW (IRENA, 2025). This remarkable growth in the PV market reflects a global shift towards renewable and sustainable energy technologies. Consequently, monitoring the performance of these systems and improving their economic and environmental returns has become increasingly important. Among the factors affecting PV system performance, dust accumulation on panel surfaces represents a significant challenge requiring automated detection methods. Data

imbalance is a common issue in machine learning applications where class distribution is uneven, with one class significantly outnumbering the others (Chen et al., 2025). This biases the model toward the majority class, thereby reducing its performance on the minority class (Albattah and Khan, 2025).

In practical applications, data imbalance frequently occurs in image classification, where minority classes are underrepresented. It significantly affects the performance of predictive models by focusing on the majority class and leading to poor generalization for the minority class. This imbalanced issue is mitigated in traditional data balancing techniques, such as undersampling the majority class or over-sampling the minority class using SMOTE; however, these methods fail to preserve the

* Corresponding author.

E-mail addresses: rajkumar@stu.jejunu.ac.kr (R. Kumar), ywkim@jejunu.ac.kr (Y.-W. Kim), ycb@jejunu.ac.kr (Y.-C. Byun).<https://doi.org/10.1016/j.egy.2026.109055>

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spatial information crucial for image-based tasks (Wang et al., 2025). Ozdemir et al (Özdemir et al., 2021). investigate SMOTE, which generates synthetic samples; it does not preserve the spatial content inherent in image data. Moreover, recent research by Shrivan et al (Hayaiean Shirvan et al., 2025). suggests that traditional methods, such as SMOTE, may not effectively capture the complex distributions of minority classes when balancing the dataset. They propose using generative models to overcome the issue of data imbalance. Based on these facts, SMOTE and Stable Diffusion are applied to measure the overall efficiency of the balanced dataset in this study. Solar panel efficiency is a critical factor in improving the performance of renewable energy systems. However, dust accumulation on solar panels is a huge challenge because it obstructs sunlight and reduces the overall efficiency of the panels. Machine learning techniques, particularly convolutional neural networks (CNN), have shown promise in automating dust detection on solar panels by using image datasets captured by drones or cameras (Onim et al., 2022). This study addresses the issue of data imbalance in the solar panel dust detection system by utilizing SMOTE and Stable Diffusion, with a focus on comparing the effectiveness of Stable Diffusion and SMOTE in balancing the dataset.

Data imbalance is one of the challenging problems in machine learning, where predictive models fail to perform better due to the uneven nature of the dataset (Miftahushudur et al., 2025). While the effectiveness of SMOTE is hamstrung by its dependence on a vectorized feature space, this feature space may not contain information critical to the performance of image-based tasks. Traditional data balancing techniques are unable to balance data with real-world synthetic samples. To address these challenges, this study presents a novel approach that combines SMOTE and Stable Diffusion to address the issue of data imbalance. In the context of solar panels, dust layers prevent sunlight from reaching the solar cells, resulting in a significant reduction in energy production. In regions with high dust on solar panels, efficiency losses of up to 20 % or more are reported (Yakubu et al., 2024). The timely detection of dust on solar panels is crucial for effective cleaning and maintenance, ensuring optimal energy generation and prolonging the lifespan of solar panels. In real-world applications, the dataset contains an unequal number of clean and dirty images, with a large number of clean panels and a small number of contaminated ones. Such an imbalanced nature of the dataset leads the model towards biases, and it tends to predict the majority class data more accurately than the minority ones (Temraz and Keane, 2022). In this study, the issue of data imbalance is addressed by employing advanced data augmentation techniques, including SMOTE and Stable Diffusion.

Dust accumulation (soiling) has a direct and measurable impact on photovoltaic (PV) system performance by reducing the effective irradiance reaching the solar cells. Field studies report that energy-yield losses vary significantly with environmental conditions and exposure duration: losses of 3–5 % have been observed annually in temperate climates, while arid and high-pollution regions such as the Middle East and South Asia experience losses exceeding 30–50 % within extended periods without cleaning (Said et al., 2024). Beyond uniform light attenuation, non-uniform dust deposition creates partial shading that forces affected cells into reverse bias, causing localized resistive heating and hotspot formation. These thermal anomalies, which can elevate cell temperatures by 10–20°C above surrounding cells, accelerate encapsulant yellowing, delamination, and long-term module degradation (Dhimish and Tyrrell, 2022). Soiling rates are influenced by site-specific factors: high humidity increases dust adhesion through capillary forces, low tilt angles reduce natural cleaning by rain and gravity, and proximity to pollution sources elevates particle deposition rates (Isaifan et al., 2019). Given this variability, automated detection of dust accumulation is essential for optimizing cleaning schedules, minimizing energy losses, and maintaining reliable long-term PV operation.

Recent studies have advanced various methods to address data imbalance in image-based environmental monitoring and fault detection tasks. Traditional approaches, such as the Synthetic Minority Over-

Sampling Technique (SMOTE), although widely used, struggle to preserve essential spatial features in image datasets. To overcome these limitations, generative methods such as Generative Adversarial Networks (GANs) and diffusion-based models have emerged, producing more realistic synthetic samples. Sousa et al. (Yilmaz et al., 2024) for instance, demonstrated the effectiveness of diffusion models in augmenting Earth observation images, while latent diffusion models significantly enhanced generalization in medical image classification. Additionally, CNN architectures such as ResNet, VGG, and MobileNet are commonly used for dust detection on solar panels due to their robust spatial feature extraction capabilities (Onim et al., 2022). Despite these advances, integrating advanced generative methods such as Stable Diffusion with CNNs for solar panel fault detection remains underexplored, representing a critical research gap addressed by this study.

This study aims to address the data imbalance issue in the solar panel dust detection dataset by applying advanced data augmentation techniques of SMOTE and Stable Diffusion. These augmentation techniques are used to balance the dataset, thereby improving the performance accuracy of CNN models (Poojary et al., 2021). The approach will enhance the model's accuracy in detecting dust, improving maintenance strategies and energy efficiency in solar panels, and contributing to the entire region through automated maintenance in renewable energy systems. The key contributions of this research can be summarized as follows:

- We propose a hybrid augmentation framework that combines feature-level oversampling (SMOTE) with image-level generative augmentation (Stable Diffusion) to address dataset imbalance in solar panel dust detection. This dual-level strategy ensures both numerical class balancing and the preservation of spatial image structures, which are often lost in classical oversampling methods.
- We demonstrate that Stable Diffusion, compared to conventional GAN-based augmentation, provides prompt-driven control and generates photorealistic synthetic images with fewer artifacts, effectively capturing diverse dust accumulation patterns on PV modules. This highlights Stable Diffusion's unique suitability for small and imbalanced PV datasets.
- We conduct a systematic evaluation across multiple CNN architectures (VGG-16, ResNet50, and MobileNetV3) using stratified 10-fold cross-validation and imbalance-sensitive metrics (F1-score, Cohen's κ , and MCC). Results show that Stable Diffusion augmentation significantly reduces false negatives and enhances sensitivity to dusty panels, outperforming SMOTE-only augmentation.
- We provide one of the first domain-specific applications of Stable Diffusion in PV dust detection, highlighting its potential to improve reliability in solar maintenance workflows. The study also establishes a methodological pathway for applying diffusion-based augmentation in other small-scale, imbalanced image classification problems.

The proposed framework not only addresses dataset imbalance but also contributes to more reliable PV system monitoring, which can support improved maintenance scheduling, optimized energy yield, and reduced operational costs in real-world solar deployments.

This paper is organized into several sections to present the proposed research for solar panel dust detection comprehensively. 2 provides a comprehensive review of the literature on dust detection and data balancing techniques, which primarily focus on the foundations of our research. 3 describes the methodology applied in this research, which consists of data collection, preprocessing, model development, limitations, and model evaluation. 4 discusses the computational environment and resources used for training and evaluation of the model. 5 presents the experimental results. 6 provides a comprehensive discussion of findings and evaluation. 7 discusses the final achievement and future directions for the research. For clarity and ease of understanding, all abbreviations and notations used in this research are summarized in

Appendix A.

2. Background

Dust detection on solar panels is a critical challenge because accumulated dust reduces energy efficiency and increases maintenance demands (Said et al., 2024). Over the years, approaches have evolved from traditional image processing techniques, which struggled under variable lighting and complex dust patterns, to advanced machine learning models capable of extracting richer spatial features. Among these, convolutional neural networks (CNNs) have consistently shown strong performance in recognizing dust patterns, demonstrating their suitability for automated PV monitoring. However, most CNN-based studies assume relatively balanced datasets, whereas real-world PV images are often highly imbalanced, with clean panels far outnumbering those with dust. This imbalance reduces sensitivity to minority classes and leads to biased predictions (Zou and Wang, 2024). To address this, researchers have explored the use of synthetic data generation with generative models. For example, diffusion models have been applied to enhance PV datasets, yielding promising improvements in fault detection accuracy (Sousa et al., 2024), while GAN-based approaches have generated diverse synthetic samples (Yang and Zhou,). Collectively, these works highlight the potential of generative augmentation but also reveal a limitation: they typically consider diffusion or GAN methods in isolation, without integrating classical oversampling techniques such as SMOTE. Our study fills this gap by providing the first comprehensive evaluation that directly compares feature-level oversampling (SMOTE) and image-level augmentation (Stable Diffusion) for balancing solar panel dust datasets, thereby offering a more holistic strategy to improve CNN performance under real-world imbalanced conditions.

2.1. Addressing data imbalance with augmentation techniques: SMOTE and stable diffusion

Class imbalance is a persistent challenge in machine learning, especially when minority classes are underrepresented during training. Several studies have explored the Synthetic Minority Oversampling Technique (SMOTE) to address this issue across different domains. For example, Alam et al. combined SMOTE with transfer learning for medical image classification. Further, Mardianta et al. applied SMOTE to

diabetic retinopathy images, further enhancing results with CLAHE-based preprocessing. Similarly, Uysal and Gao et al. demonstrated that incorporating SMOTE, sometimes coupled with contrastive learning, improves robustness and accuracy in imbalanced datasets. Collectively, these studies confirm that SMOTE is effective in generating synthetic samples that will enhance classification performance. However, SMOTE and its variants (e.g., DeepSMOTE (Dablain et al., 2022)) primarily operate in feature space, which means they overlook spatial structures that are critical in image-based tasks such as PV panel monitoring. This limitation has prompted researchers to explore generative models, such as GANs and diffusion approaches, which can produce realistic synthetic images that better preserve spatial information. For instance, Strelcenia and Prakoonwit (Strelcenia and Prakoonwit, 2023) highlighted GANs as practical tools for addressing imbalance in financial fraud detection. In contrast, Hasib et al. identified SMOTE as one of the most reliable techniques, acknowledging the growing role of generative methods. Taken together, the literature shows that SMOTE-based oversampling remains a cornerstone technique for handling imbalance, yet it is constrained when applied to complex visual datasets. In contrast, generative augmentation methods, such as GANs and diffusion models, offer more realistic data synthesis but are often studied in isolation. This motivates our approach of directly comparing SMOTE and Stable Diffusion, thereby bridging the gap between feature-level oversampling and image-level generative augmentation for solar panel dust detection.

Table 1 highlights that most PV-specific studies rely on CNN-based methods combined with oversampling (e.g., SMOTE), while generative approaches such as Stable Diffusion are emerging but largely untested in PV contexts. Notably, none of the studies systematically benchmark feature-level and image-level augmentation on solar panel dust datasets, which defines the gap addressed in our research.

2.2. Generative models for synthetic data generation in low data scenarios

Generative models have emerged as powerful tools for addressing data scarcity and imbalance by producing synthetic samples that closely resemble real-world conditions. In low data settings, diffusion models have shown particular promise. For example, Sagers et al., n.d used latent diffusion models to improve medical image classification. In contrast, Zhou et al., n.d., and Rahat et al. (2024) emphasized the ability

Table 1
Previous research work on imbalance datasets.

Ref	Year	Dataset	Augmentation / Models Used	Key Findings	Limitations
(Sefer and Kaya, 2024)	2024	Solar Panel Images	Deep Learning (CNN)	Proposed a deep learning model for detecting dust accumulation on solar panels.	Lack of specific performance metrics.
(Uysal, n.d.)	2025	Photovoltaic Panel Images	CNN-based Deep Learning Models	Introduced a CNN-based method for detecting dust accumulation on solar panels in optoelectronics.	Limited dataset description.
(He et al., 2024)	2024	Photovoltaic Solar Panel Images	Improved MobileNet Algorithm	Developed an improved MobileNet-based dust detection system.	Limited evaluation of accuracy.
(Gong and Ma, 2024)	2024	Solar Panel Surface Images	Deep Learning	Applied deep learning for detecting dust on solar panel surfaces.	No reported accuracy or dataset statistics.
(Mehta and Saini, 2024)	2024	Solar Panel Dust Dataset	CNN-GAN	Used CNN and GAN for dust detection and classification.	High computational cost.
(Du et al., 2025)	2025	MNIST	SMOTE + Various ML Models	Demonstrated effective hierarchical feature extraction in image classification.	Limited generalizability to storm detection or PV imagery.
(Bassil et al., 2024)	2024	Solar Panel Images	CNN Models + SMOTE	Applied SMOTE with CNN models for dust detection.	Lack of comparison with other architectures.
(Rai, Kinzang and Choden, 2024)	2024	Solar PV Panel Images	SMOTE + CNN	Proposed a cost-effective CNN-based dust detection solution.	Limited to a small dataset.
(Sagers et al.,)	2023	Medical Image Dataset	Latent Diffusion + CNN	Used latent diffusion for image augmentation in medical imaging.	Focused only on medical images.
(Sousa et al., 2024)	2024	Earth Observation Image Dataset	Diffusion Model Augmentation	Implemented diffusion models for data augmentation in Earth observation tasks.	Limited to Earth observation domain.
(Yilmaz et al., 2024)	2024	Industrial Area Images	Stable Diffusion + LoRA	Generated synthetic industrial images using Stable Diffusion with LoRA.	Limited to industrial datasets.
(Benkedadra et al., 2024)	2024	Image Dataset	Stable Diffusion (CIA Framework)	Proposed controllable image augmentation (CIA) framework based on Stable Diffusion.	Limited evaluation with diverse models.

of generative AI to expand training sets and enhance generalization. Collectively, these studies demonstrate that generative models can substantially improve classification performance by increasing the diversity of training data. Recent work has extended these ideas into environmental and industrial domains. Sousa et al. (2024) demonstrated that diffusion-based augmentation improves Earth observation data and is even suitable for detecting solar panel dust. In contrast, Yilmaz et al. (2024) and Benkedadra et al. (2024) demonstrated the flexibility of Stable Diffusion for generating realistic synthetic images in industrial and cross-domain contexts. Together, these findings highlight Stable Diffusion as a versatile augmentation tool capable of preserving spatial realism while addressing imbalance. Despite these advances, most studies apply diffusion or other generative models in isolation, without comparing them against classical oversampling approaches such as SMOTE. This limits our understanding of whether feature-level or image-level augmentation provides greater benefit for PV dust detection. To address this gap, our work systematically benchmarks Stable Diffusion alongside SMOTE, offering a more comprehensive perspective on dataset balancing strategies.

2.3. Solar panel dust detection techniques: challenges and solutions

Dust accumulation on solar panels can significantly degrade performance, making accurate detection essential for efficient maintenance and energy output. Early studies relied on traditional image processing techniques such as edge detection and thresholding, but these methods often failed under variable lighting and complex dust patterns (Farah et al., 2022). The shift to deep learning has improved robustness in such conditions. Safer and Kaya Sousa et al. (2024) and Uysal, n.d, demonstrated that convolutional neural networks (CNNs) outperform classical methods by automatically extracting discriminative features from raw images. He et al. (2024). further enhanced this line of work by adapting MobileNet for PV dust detection, showing that lightweight CNNs can maintain accuracy while remaining suitable for edge deployment. Hybrid models that integrate CNNs with other machine learning techniques have also been explored. For example, Najafli and Imamverdiyev (2023) and Mehta and Saini (2024) combined CNNs with SVMs to enhance predictive performance and directly link dust detection to energy output optimization. Similarly, Gong and Ma (2024) emphasized that CNN-based methods are particularly effective in handling environmental variability in real-world conditions. Collectively, these studies highlight CNNs as the state-of-the-art approach for solar panel dust detection, offering higher accuracy and generalization compared to traditional methods. However, a recurring limitation is that CNN performance remains sensitive to class imbalance, where minority cases (e. g., heavily dusted panels) are underrepresented. This underscores the need to combine CNN-based models with advanced augmentation strategies that can both balance datasets and preserve spatial realism, an approach our study systematically investigates.

2.4. Convolutional neural networks in environmental monitoring: applications and advancements

Convolutional neural networks (CNNs) have become the foundation of modern image recognition because of their ability to extract hierarchical features from raw pixel data automatically. In the context of solar energy, CNNs have been widely adopted for dust detection due to their capability to capture subtle accumulation patterns that directly affect panel efficiency. Several studies have demonstrated this effectiveness: for example, Mehta and Saini (2024) combined CNN feature extraction with GAN-based augmentation to achieve high dust detection accuracy, while Onim et al. (2022). developed SolNet, a CNN framework tailored for PV dust detection under diverse environmental conditions. Similarly, Bassil et al. (2024) and Rai Kinzang and Choden (2024). highlighted the generalization ability and cost-effectiveness of CNNs, reinforcing their value for practical solar panel monitoring. Ultimately, these works

demonstrate how CNNs offer robust, accurate, and scalable solutions for environmental monitoring tasks, particularly in dust detection. However, their performance is highly dependent on the quality and balance of training data. Without appropriate augmentation, CNN-based models often struggle with underrepresented cases such as heavily dusted panels, leading to biased predictions. This dependency highlights the importance of integrating CNNs with advanced data-balancing strategies, such as generative augmentation, to ensure reliable performance in real-world PV monitoring scenarios.

2.5. Research gap and findings

The existing literature on solar panel dust detection has primarily focused on CNN-based classification and classical oversampling techniques such as SMOTE to mitigate data imbalance. While SMOTE has proven effective in generating synthetic samples, it operates in feature space and cannot capture the spatial realism needed for robust image-based applications (Gholampour, 2024). More recent efforts have explored generative approaches such as GANs and diffusion models, which can produce high-quality synthetic images and enhance generalization in low-data or imbalanced scenarios (Yilmaz et al., 2024; Sousa et al., 2024; Yang and Zhou). However, these works have typically applied generative augmentation in isolation and have not systematically compared their benefits with established oversampling methods for PV imagery. This gap is significant for PV dust detection, where fine-grained surface variations and real-world operational biases require augmentation strategies that balance classes while preserving spatial authenticity. To address this, our study provides the first comprehensive benchmarking of feature-level oversampling (SMOTE and DeepSMOTE) against image-level augmentation using Stable Diffusion, specifically for solar panel dust datasets. By evaluating these approaches under real-world imbalanced conditions, we demonstrate how generative diffusion can complement or outperform traditional methods, offering a more robust pathway toward reliable PV dust detection.

3. Methodology

The proposed methodology comprises a comprehensive pipeline to address the problem of class imbalance in solar panel dust detection using CNN models, as illustrated in Fig. 1. In this study, SMOTE is employed to address the class imbalance in the solar panel dust detection dataset by oversampling the minority class, thereby creating a balanced dataset for training. However, traditional SMOTE does not preserve the spatial features of image data. To overcome this, realistic, high-quality synthetic images of dusty panels are generated using Stable Diffusion, as spatial relationships between CNNs used in model training (Pavez et al., 2023). Furthermore, ResNet50, VGG-16, and MobileNetV3 were chosen based on their proven performance in image classification tasks. ResNet50 residual learning enables the deep networks to learn complex features without the vanishing gradient problem (Qawasmeh et al., 2025). VGG-16, with its deep architecture, excels in extracting features from image data. Moreover, MobileNetV3 is efficient in real-time applications with limited computational resources. The comprehensive approach applied in this research is illustrated in Fig. 2. Data augmentation techniques are then used to mitigate the dataset's imbalance, which is typically characterized by a large number of clean panel images compared to unclean ones. SMOTE and Stable Diffusion are applied in two approaches: SMOTE, which synthetically interpolates new minority samples, and Stable Diffusion, which is used for generating realistic images of presentations with dust. First, a balanced dataset is used to train CNN models, including VGG-16, ResNet50, and MobileNetV3, for dust classification. In the last part, metrics used to evaluate the model include accuracy, precision, recall, and F1-score, which measure its effectiveness in relation to each technique of balancing, technique, and architecture employed. Key assumptions, limitations, and sources of uncertainty are summarized in 6.14.

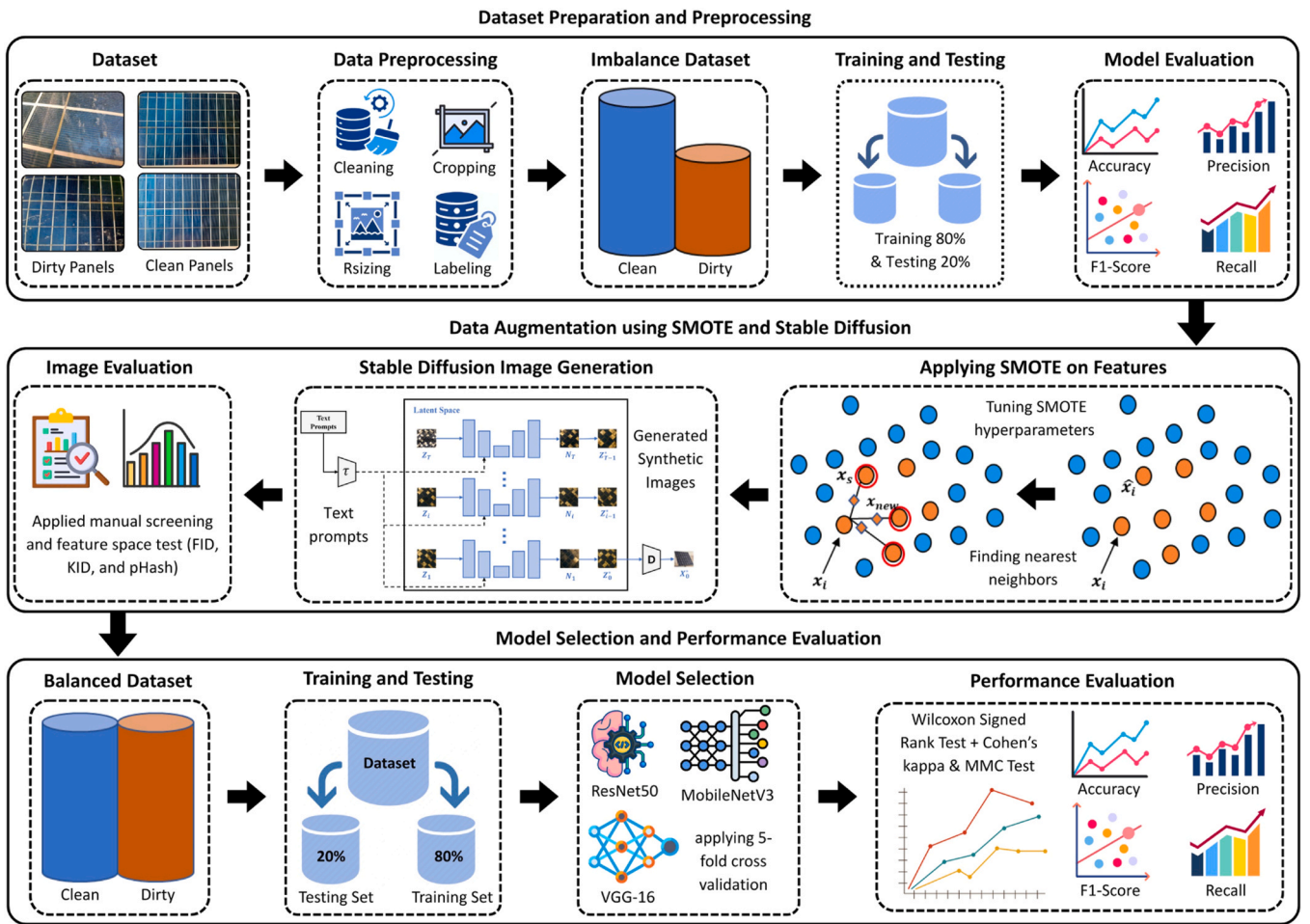


Fig. 1. Proposed methodology: dataset imbalance was addressed using SMOTE and Stable Diffusion with validation (manual + FID/KID/pHash), followed by CNN training and evaluation using standard metrics and significance testing.

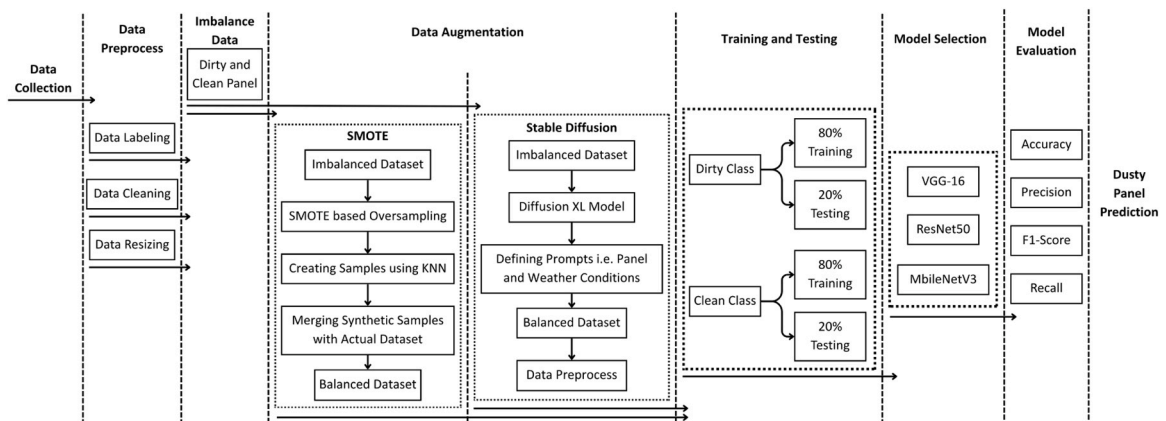


Fig. 2. Proposed methodology workflow: data were preprocessed and balanced using SMOTE and Stable Diffusion, then split into training and testing sets. Models (VGG-16, ResNet50, MobileNetV3) were evaluated with accuracy, precision, recall, and F1-score for reliable classification.

3.1. Dataset description

This study primarily addresses the dataset imbalance issue using SMOTE and Stable Diffusion techniques in conjunction with CNN models, making this proposed approach applicable in various fields. However, in this research, the imbalanced solar panel dust detection dataset, as shown in Fig. 3, is used to evaluate the performance of CNN models on both balanced and imbalanced datasets. The dataset was

collected from various regions in Bangladesh and is publicly available as the SolNET dataset (Onim et al., 2022). For model training, all images were processed as RGB inputs and resized to $224 \times 224 \times 3$ pixels. The images were obtained from the publicly available SolNET dataset (Hayaian Shirvan et al., 2025).

The SolNET images were acquired in real outdoor environments and therefore exhibit natural variability in brightness, illumination, shadow presence, viewing angle, and background context. However, acquisition

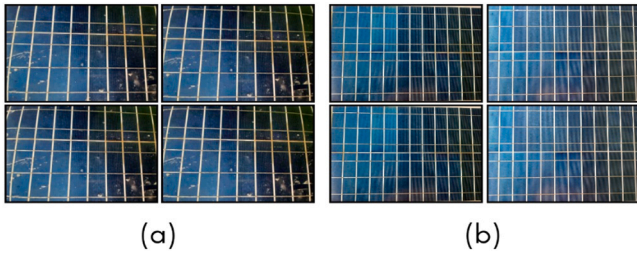


Fig. 3. (a) Dirty Panels and (b) Clean Panels.

metadata such as ambient temperature, irradiance level, camera-to-panel distance, and capture time are not provided in the dataset release; consequently, these factors could not be controlled or parameterized in this study. The dataset release does not report camera model, sensor specifications, or the original capture resolution (pixel count) of the images. Therefore, in this study we report the effective resolution used for training: all samples were processed as RGB images and resized to 224×224 pixels to match the CNN input requirements. Resizing ensures uniformity for CNN model input, although it may cause a minor loss of fine details, such as dust particle texture (Saponara and Elhanashi, 2021). Since resizing was applied consistently to all samples, the impact on comparative model evaluation is minimal. The dataset comprises images of photovoltaic (PV) panels captured under different environmental and lighting conditions with the intention of classifying images into two distinct panel categories: clean panels and dusty panels, as shown in Fig. 3. These images correspond to polycrystalline silicon (poly-Si) PV modules, as specified in the SolNET dataset (Onim et al., 2022) description. It should be noted that the SolNET dataset consists exclusively of polycrystalline silicon (poly-Si) PV modules. Other PV technologies, such as monocrystalline or thin-film modules, which have different textures and appearances, were not included in this dataset. Specifically, the dataset comprises 502 images of clean solar panels and 340 images of panels with dust. The class distribution in this case suggests a moderate class imbalance, where clean panel images significantly outnumber the dirty panel samples. Such an imbalanced dataset can lead to biased learning in deep models, as deep CNN architectures tend to be biased toward the majority class (Ghosh et al., 2024). This results in becoming less sensitive towards the minority class, which is critical in context, since identifying dust accumulation is the main objective of this research study. Thus, it is essential to balance the dataset by employing more advanced augmentation techniques, such as SMOTE and Stable Diffusion, which will enhance the model's generalization ability and predictive performance.

3.2. Sample size adequacy

The SolNET dataset used in this study is relatively modest (502 clean and 340 dusty images), it was selected because it represents one of the few publicly available collections specifically focusing on polycrystalline silicon PV panels under real-world dust conditions. Collecting larger datasets is logistically challenging due to seasonal and environmental dependencies. To ensure the reliability of the findings despite this constraint, we employed 10-fold cross-validation, reported multiple performance metrics (accuracy, κ , MCC, and F1-score), and evaluated the results across several CNN backbones. These methodological safeguards provide a robust basis for comparing augmentation strategies under conditions where data scarcity is a common and realistic challenge in PV inspection tasks. In our future work, we plan to extend this approach to larger, real-time datasets collected from multiple sites, further enhancing generalizability.

3.3. Data pre-processing techniques

To ensure consistency and improve model performance, several data

preprocessing steps were applied to the solar panel dust detection dataset. As illustrated in the workflow (Fig. 1), the initial raw images were cleaned, removing corrupted or low-quality samples, so that the input data is free of noise and visually unified. This was then followed by resizing all the images to a standard dimension of 224×224 pixels, suitable for CNN architectures, to ensure the input shape across the dataset remains uniform. While resizing ensures compatibility across different CNN models, it may also result in a slight loss of fine-grained details such as small dust particles or subtle surface textures (Saponara and Elhanashi, 2021). However, since resizing was applied consistently to all images, the effect on comparative evaluation was minimal and did not prevent the models from achieving high performance. Then, to refine the model, it proceeded with cropping, where the panel area was given direction, and irrelevant backgrounds or borders that could confuse the model were removed. The last was to manually label each image as both “clean” and “dusty” for supervised training of the classification task. For the SMOTE-based augmentation, numerical preprocessing was needed. First, all images were converted to the RGB format and then normalized into the $[0, 1]$ range to stabilize learning (Pei et al., 2023). To make SMOTE applicable for this baseline comparison, images were reshaped into feature vectors. While this ignores spatial correlations, it allows the use of classical SMOTE techniques for preliminary benchmarking. For comparison purposes, SMOTE was applied in the feature vector space after resizing and normalizing images. While this omits spatial structure, the goal was to evaluate its baseline effect on class balancing in image classification (Rashid et al.,). Thus, the synthetic samples from SMOTE were created to represent valid changes in the minority class. In the case of Stable Diffusion, text prompts and latent representations were used to synthesize the synthetic images. The generated images were cropped to retain only the most significant solar panel regions, without introducing additional artifacts or irrelevant backgrounds. The images were then cropped to fit the original dataset's height and width, maintaining uniformity for the model training. The preprocessing strategies ensure that high-quality, standardized, and semantically relevant inputs are provided to the model for both real and synthetic datasets, enabling the detection of dust accumulated on solar panels (Balachandran et al., 2024). In summary, while image resizing to 224×224 pixels may slightly reduce fine-grained dust texture information, applying the same resizing procedure to all real and synthetic samples ensures fair comparison across augmentation strategies and does not affect the relative performance conclusions under this fixed input-resolution setting.

3.3.1. Justification and impact of geometric data augmentation transformations

To make the dataset more diverse and effective, traditional geometric augmentation transformations are applied, as the original dataset is small and imbalanced, which makes it risky to overfit and leads to poor generalization. To address the limited size and class imbalance in the original dataset (502 clean, 340 dusty images), first applied geometric augmentation shown in Table 2 using random rotations (up to 40°), width and height shifts (up to 20 %), shear (up to 20°), zoom (up to 20 %), and horizontal flips. These transformations generate additional image variations in orientation, position, scale, perspective, and mirroring. This expansion of data diversity reduces overfitting and improves model generalization to real-world solar panel conditions. Nonetheless,

Table 2
Variations applied for augmented images.

Augmentation	Variation Details
Rotation	Random rotations (e.g., $\pm 40^\circ$)
Shift	Horizontal/Vertical shifts (e.g., $\pm 20\%$)
Shear	Shear transformations (e.g., $\pm 20^\circ$)
Zoom	Random zoom (in/out up to 20 %)
Horizontal Flip	Random mirroring (left/right)

geometric augmentation alone does not resolve class imbalance, so further SMOTE and Stable Diffusion are applied in later stages. This helps CNN models learn features that are robust to such changes, improving accuracy on unseen data.

3.4. Data augmentation to address imbalanced datasets

3.4.1. Synthetic minority oversampling technique (SMOTE)

The Synthetic Minority Oversampling Technique (SMOTE) is a popular method for addressing the class imbalance in datasets by generating synthetic samples for the minority class (Elreedy and Atiya, 2019). It works by selecting random samples from the minority class and creating new samples based on the feature vector space that are combinations of the selected sample and its nearest neighbors (Li et al., 2021). Fig. 4 illustrates the SMOTE-based augmentation workflow used in this study as a baseline approach to address the class imbalance. The minority class samples are abstracted in feature space, as illustrated in Fig. 4. SMOTE is typically designed for tabular data, but it can also transform image data into vector representations to facilitate comparison and analysis.

However, this reshaping process sacrifices the spatial relationships that are critical for image-based tasks, such as dust detection on solar panels. This limitation is acknowledged in the study, and Stable Diffusion is used to address the spatial integrity issues. Nevertheless, SMOTE was included as a classical oversampling technique to serve as a performance baseline, with its limitations in image-based tasks explicitly recognized. Before applying SMOTE, the dataset was preprocessed by resizing, converting to RGB, and normalizing the data. To enable its use, image data were reshaped into vectorized form, which inevitably discards spatial correlations but permits the application of traditional feature-level oversampling. SMOTE mitigates this by selecting a minority class instance, identifying its k nearest neighbors, and generating

synthetic samples through linear interpolation between the selected instance and its neighbors (Mukherjee and Khushi, 2021). These interpolated samples enhance the representation of the minority (dusty panel) class, providing a more balanced dataset for subsequent model training. Once the neighbors are identified, they are then used to generate new synthetic samples by interpolating between the selected sample and its neighbors, thereby creating realistic variations of similar panels with dust (Mukherjee and Khushi, 2021). The newly generated samples, x_{new} , help increase the minority class representation, as shown in Eq. 1.

$$x_{new} = x_i + \delta(\hat{x}_i - x_i) \quad (1)$$

where:

x_{new} is the synthetic sample,

$\hat{x}_i$ is a randomly selected neighbor of x_i .

δ is a random number between 0 and 1 that controls the interpolation.

It should be noted that SMOTE was included in this study only as a classical baseline to provide a reference point for comparison. While reshaping image data into feature vectors discards important spatial information, making SMOTE less effective for visual tasks, its inclusion highlights the relative advantages of image-level augmentation achieved through Stable Diffusion.

3.4.2. Stable diffusion based synthetic image generation

Stable Diffusion is a generative model that utilizes denoising diffusion probabilistic models (DDPMs) to generate high-quality synthetic images from random noise (Müller-Franzes et al., 2023). Unlike traditional methods such as SMOTE, Stable Diffusion is the most appropriate method for balancing the dataset because it produces high-quality, realistic synthetic data, ensuring diversity and improving the model's

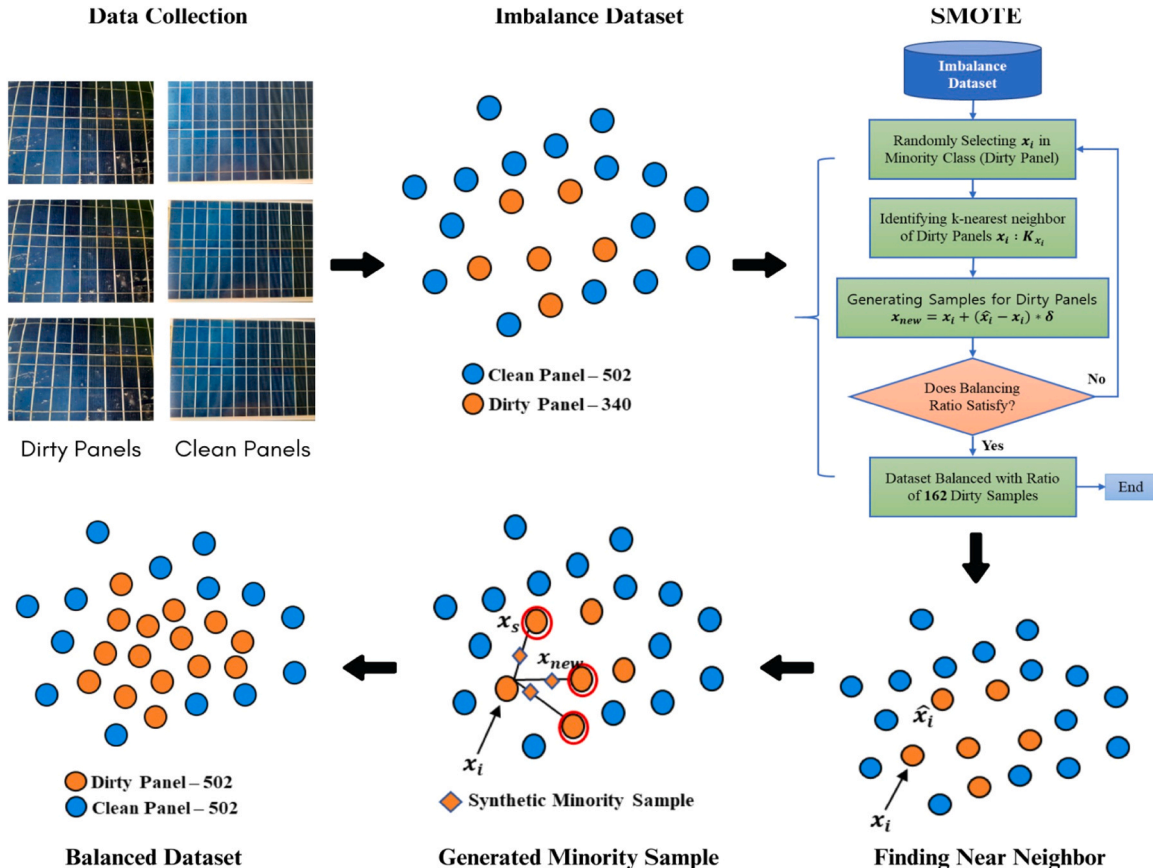


Fig. 4. Baseline SMOTE application: minority class sample generation by reshaping images into feature vectors.

predictive accuracy for dust detection on solar panels (Stöckl and n.d., 2023). In this study, Stable Diffusion was utilized to generate diverse and high-quality images of dusty solar panels, thereby enhancing model generalization by mimicking real-world dust patterns. The process begins with an initial random noise image Z_0 , which is progressively transformed into a structured latent space as illustrated in Fig. 5. This transformation is achieved by encoding the input image X_0 into the latent space using an encoder (denoted by Z_0 in Fig. 5), producing the latent representation Z_0 . The model then performs forward propagation through a series of denoising steps, where the noisy image is refined iteratively to resemble the realistic features of a solar panel with dust. The diffusion process works by adding noise to the input image in several steps and reversing this process using a U-Net decoder (Chowdary and Yin, 2023). For each step, the model predicts the noise in the image and refines it. The Eq. (2) for this process is as follows:

$$Z_t = Z_{t-1} + Z_{(t-1)NoisePredictions} \quad (2)$$

where Z_t is the noise image at step t , and $Z_{(t-1)NoisePredictions}$ is the predicted noise used to refine the image. The input image X_0 is first encoded using an Encoder (E) into a latent space representation Z_0 . The forward diffusion process adds noise at each step, creating progressively noisier images $Z_1, Z_2, \dots, Z_t$ as given in Fig. 5. A Denoising U-Net then reverses this process in the reverse diffusion phase, reconstructing the image from Z_t back to a clean solar panel image X_0 . The dirty panel images generated through Stable Diffusion are shown in Fig. 6. To generate synthetic dust images, text prompts are initially given through a Stable Diffusion model based on the conditions (e.g., “Dust Accumulation” or “Bright Sunlight”) that are passed through the latent space. In the forward diffusion process, as illustrated in Fig. 5, a clean image of the solar panel Z_0 is corrupted with noise in the latent space, with each image at timestep Z_i becoming progressively less noisy, and the final timestep Z_T being pure Gaussian noise. Finally, the denoised representation Z^*0 is decoded into the pixel space (i.e., the final generated image using Decoder D).

3.4.3. Stable diffusion augmentation protocol

To mitigate class imbalance, synthetic images of dusty panels were generated using Stable Diffusion XL (SDXL) with the checkpoint stabilityai/stable-diffusion-xl-base-1.0, implemented via the Hugging-Face Diffusers library (v0.24.0) (Stability, 2023). Generations were performed on an NVIDIA RTX 4060 GPU in float16 precision with attention slicing enabled for memory efficiency.



Fig. 6. Synthetic images of dirty solar panels generated using Stable Diffusion.

- **Sampling Strategy:** The sampling process employed the DDIM scheduler with 30 denoising steps, a guidance scale of 7.5, and a base resolution of 1024×1024 , subsequently resized to meet the input requirements of each CNN model. A total of 162 augmented images were accepted following the filtering procedure. The consolidated configuration settings for Stable Diffusion augmentation are summarized in Table 3.
- **Prompt Design:** To capture realistic diversity in the minority (dusty) class, carefully designed prompts were applied. Dust-related prompts included bird droppings, dust accumulation, moss growth, dark spots, and a heavy dust layer, while weather-related prompts involved clear sky, bright sunlight, heat conditions, sunset with shadows, and partly cloudy weather. These variations ensured both surface-level dust representation and environmental context variability. The complete list of prompt variations is presented in Table 3.
- **Filtering and Acceptance:** All generated images were subjected to a two-step filtering process. First, manual vetting eliminated outputs with distorted solar panel geometry, text artifacts, or irrelevant

Table 3

Stable Diffusion augmentation configuration (consolidated).

Parameters	Setting
Checkpoint	stabilityai/stable-diffusion-xl-base-1.0
Library/Version	Diffusers v0.24.0
Scheduler	/ Steps DDIM / 30
Guidance Scale	7.5
Base Resolution	1024×1024 (resized per model)
Precision / Memory	float16, attention slicing
Hardware	RTX 4060
Accepted Images	162 (see 6.11 for timings)
Leakage Controls	Train-fold only; no val/test conditioning; pHash filtering
Artifacts Filter	Manual vetting + rejection list

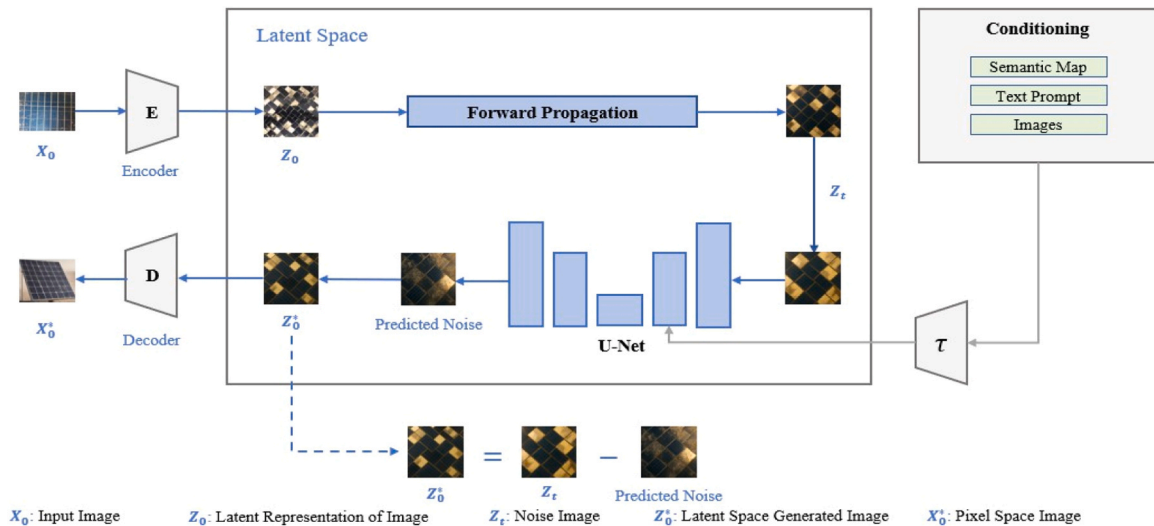


Fig. 5. Stable Diffusion-based synthetic image generation for minority class balancing.

backgrounds. Second, near-duplicate images were detected and removed using perceptual hashing to ensure diversity. This rigorous filtering resulted in 162 high-quality synthetic images used for training.

- **Leakage Prevention:** To avoid any possibility of data leakage, augmentation was applied strictly within training folds only. Validation and test images were never used for conditioning, and duplicate checks ensured that no synthetic image overlapped with evaluation data. Fixed seeds were used per fold in the main experiments, and additional runs with varied seeds confirmed the robustness of the results.

3.5. Validation of synthetic data generated through stable diffusion

To ensure the reliability and realism of the images generated by Stable Diffusion, we employed a two-fold validation strategy. First, visual quality and artifact screening were performed using domain-specific criteria to confirm geometric integrity, realism of dust patterns, consistency of lighting, and artifact-free representation. Second, embedding similarity tests were conducted in the feature space using quantitative metrics, including FID, KID, perceptual hashing, and distributional histograms. Together, these complementary approaches ensured that only high-quality and statistically consistent images were retained.

3.5.1. Visual quality and artifact screening

All generated images underwent a systematic vetting process to ensure realism and domain accuracy. A total of 300 synthetic images were initially produced. Each image was then subjected to a two-step evaluation. First, manual screening was conducted independently by two evaluators, who compared the synthetic outputs with real-world PV panel images to assess their plausibility. The screening criteria included (see Table 4): (i) integrity of panel geometry and module structure, (ii) natural distribution of dust and soiling patterns, (iii) visual consistency with realistic weather and lighting conditions, and (iv) absence of visual artifacts such as text overlays, unnatural textures, or distorted shapes. Second, perceptual hashing was applied to detect and eliminate near-duplicate samples, ensuring diversity in the retained set. Following this rigorous process, 162 images were selected for use in augmentation, while the remaining samples were discarded due to unrealistic features or redundancy. This careful evaluation ensured that only high-quality synthetic images, comparable to real-world scenarios, were included in the final dataset.

3.5.2. Embedding similarity (Feature Space Test)

To ensure that the synthetic images generated by Stable Diffusion

were both realistic and diverse, we employed three widely used validation metrics: Fréchet Inception Distance (FID), Kernel Inception Distance (KID), and perceptual hashing (pHash). FID measures the distributional similarity of synthetic images to real images in feature space, where lower values indicate closer resemblance to the real dataset (Khosravi et al.,). KID provides an unbiased alternative to FID, particularly useful for smaller datasets, by quantifying distributional alignment via polynomial kernel maximum mean discrepancy (Jayasumana et al., 2024). Perceptual hashing (pHash), on the other hand, evaluates near-duplicate rates between real and synthetic images, ensuring that generated samples add novel diversity rather than replicating existing images (Morales et al.,). In our experiments, the baseline Stable Diffusion set achieved an FID of 107.65 and KID of 0.0455, reflecting a clear domain gap. After applying HSV calibration, FID and KID improved to 87.09 and 0.0295, respectively, as shown in Table 5. Finally, filtering reduced the set to 162 high-quality synthetic images, yielding an FID of 63.86 and KID of 0.0220. These results demonstrate a progressive improvement in statistical similarity between real and synthetic images. Importantly, pHash analysis confirmed 0 % duplication, ensuring diversity.

Together, these metrics validate the use of our synthetic dataset for balancing PV panel defect classes. Further support comes from our histogram-based statistical analyses (Fig. 7), which demonstrate a substantial overlap between the real and synthetic distributions in terms of intensity standard deviation and saturation. This indicates that the generated images preserve essential texture and contrast characteristics of real PV defect images. Taken together, these results show that although the FID score suggests room for improvement, the synthetic data nonetheless introduces sufficient diversity and statistical consistency to address class imbalance, thereby improving the robustness of downstream CNN models.

3.6. CNN models for dust detection

3.6.1. VGG-16

VGG-16 is a widely used deep convolutional neural network (CNN) for image classification tasks (Tammina, 2019). The model employs a sequence of small 3×3 convolutional filters and pooling operations, enabling it to capture both fine-grained and high-level image features efficiently as illustrated in Fig. 8. In this study, VGG-16 was used with input images resized to $224 \times 224 \times 3$ pixels. The model was selected due to its proven performance in various computer vision applications and its strong capability to learn complex visual patterns, which is particularly relevant for detecting subtle dust accumulation on solar panels under diverse environmental conditions (Dantas et al., 2020).

3.6.2. ResNet50

Residual Networks (ResNets) are deep CNN architectures designed to address the vanishing gradient problem by introducing residual (shortcut) connections that allow gradients to flow more effectively during backpropagation (Borawar and Kaur, 2023). ResNet50, a 50-layer variant widely adopted in computer vision tasks, consists of stacked convolutional layers, batch normalization, ReLU activations, and residual blocks that enable efficient training of very deep models (Khorra and Singh, 2025). As illustrated in Fig. 9, the architecture combines residual connections with standard convolutional layers to capture complex hierarchical features.

Table 5

Validation metrics for Stable Diffusion-generated images compared with real PV panel images.

Setting	FID ↓	KID ($\times 1e-3$) ↓	pHash Dup. (%)
Baseline (Unfiltered)	107.653	45.547 ± 9.013	0.00
After Calibration	87.085	29.507 ± 4.624	0.00
Calibrated + Filtered (162)	63.862	21.995 ± 3.879	0.00

Table 4
Evaluation rubric for filtering Stable Diffusion-generated images.

Criterion	Description	Examples of Rejection
Panel Geometry Integrity	PV modules should maintain correct shape, alignment, and structure	Warped panels, bent edges, unnatural module outlines.
Dust/Soiling Pattern Realism	Dust distribution should appear natural, varied, and comparable to real-world scenarios.	Unrealistically uniform dust layer, unnatural streaks, dust floating off-surface.
Lighting and Weather Consistency	Scene illumination and weather conditions should be plausible and match the descriptions provided in the prompts.	Shadows cast in inconsistent directions, overexposed panels, unrealistic color tones.
Artifact-Free Image	Images should be free from generative artifacts or irrelevant objects.	Text overlays, random symbols, mosaic textures, background distortions.
Diversity and Uniqueness	Each image should add new variation and not duplicate existing samples.	Duplicate or near-duplicate images detected by perceptual hashing

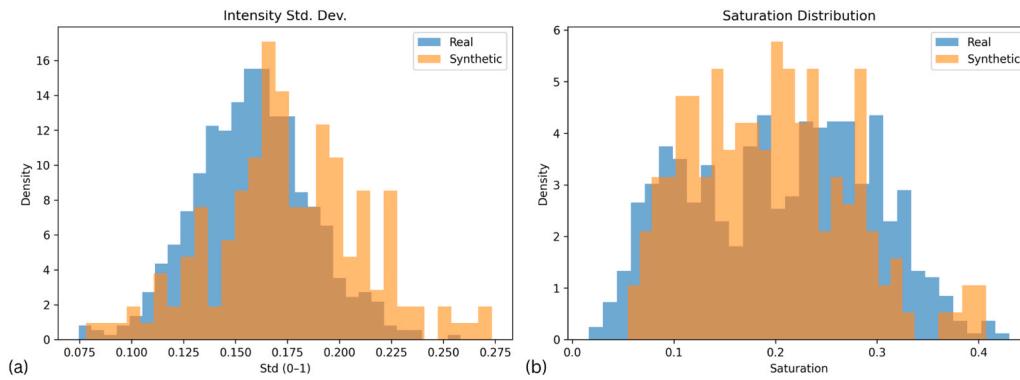


Fig. 7. Comparative histograms of real and Stable Diffusion-generated images. (a) Intensity standard deviation showing consistency in contrast and texture. (b) Saturation values indicating alignment in color richness with real PV panels.

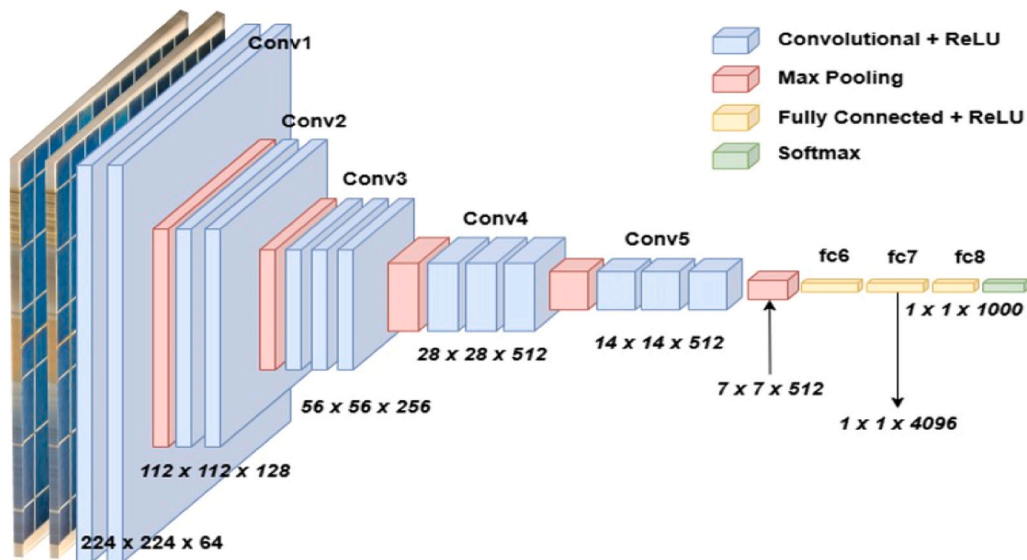


Fig. 8. VGG-16 Architecture: The model receives $224 \times 224 \times 3$ RGB images of solar panels as input and processes them through multiple convolutional, pooling, and fully connected layers to classify panels as clean or dusty.

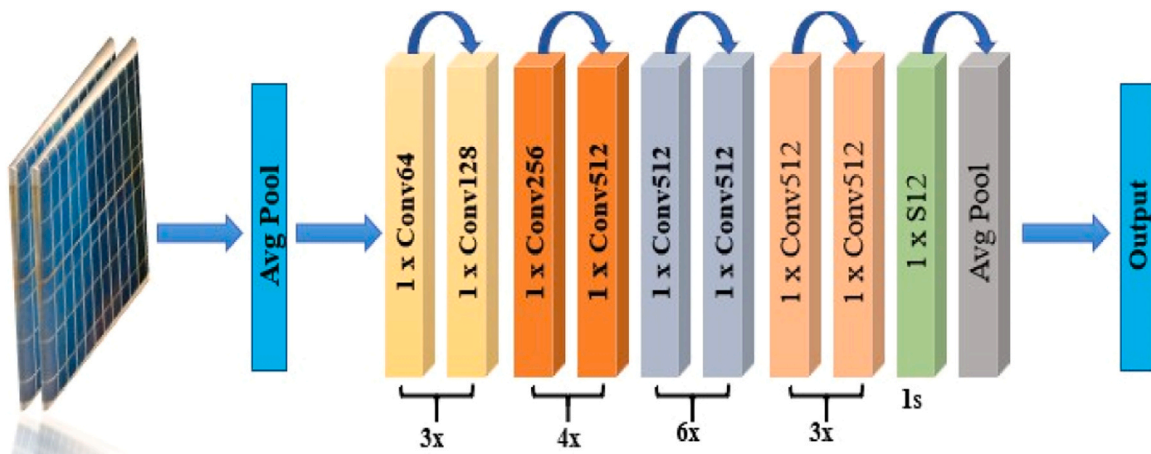


Fig. 9. ResNet50 Architecture: The model takes an input image of a solar panel and processes it through sequential convolutional layers and residual blocks, utilizing skip connections to enhance feature learning. Average pooling layers are used before the final output layer to perform the clean/dirty classification.

In this study, ResNet50 was employed for solar panel dust detection due to its ability to learn deep representations of subtle dust patterns that simpler CNNs may fail to capture, making it suitable for imbalanced

datasets where discriminating fine details is critical (Li and Wang,).

3.6.3. MobileNetV3

MobileNetV3 is a state-of-the-art lightweight CNN architecture designed for mobile and edge devices with limited computational resources (Zhao and Wang, 2022). Its efficiency is achieved through depthwise separable convolutions, lightweight attention mechanisms such as the Squeeze-and-Excitation block, and neural architecture search (NAS) for design optimization (Guo et al., 2024). As illustrated in Fig. 10, MobileNetV3 achieves a balance between accuracy and efficiency, making it suitable for real-time applications in resource-constrained environments. In this study, MobileNetV3 was employed to evaluate its effectiveness in detecting dust on solar panels under imbalanced data conditions. Its compact design enables faster training and inference while maintaining strong feature extraction capabilities, which is particularly advantageous for scalable deployment in solar monitoring systems.

3.6.4. Justification for using CNN models

Convolutional Neural Networks (CNNs) are employed for dust detection on solar panels due to their demonstrated capability in learning complex spatial patterns from image data (Kamal et al., 2019). Unlike traditional machine learning methods that require handcrafted features, CNNs automatically extract hierarchical representations directly from raw pixel information, capturing both local textures and global contextual cues relevant to dust, dirt, and surface anomalies. The effectiveness of CNNs in image-based defect classification has been validated across diverse domains, including industrial inspection and environmental monitoring, where subtle visual differences have a critical impact on detection accuracy (Wang et al., 2020). Furthermore, CNNs provide robust generalization to real-world conditions, such as variations in illumination, weather, and dust morphology, which are common in solar panel installations. These properties make CNNs highly suitable and widely adopted for visual inspection tasks requiring precise, scalable, and automated solutions.

4. Experimental setup

This section outlines the experimental setup for the research. The computational environment and resources used for model training and evaluation are described, along with the methodology for splitting the dataset into training and testing sets. Additionally, the performance evaluation metrics applied in the study are detailed. These aspects ensure reproducibility and provide a fair assessment of the proposed approach's effectiveness in detecting solar panel dust.

4.1. Computational environment and resources

The research experiment and evaluation are conducted using a GPU-enabled system, which utilizes TensorFlow and Keras for model training.

Details are provided in Table 6. The training is initially performed over 50 epochs on both an imbalanced and a balanced dataset using SMOTE. However, for the Stable Diffusion-based balanced dataset, 80 epochs are performed for each model. The software stack includes Python libraries such as Numpy, Pandas, matplotlib, and TensorFlow, with additional use of SMOTE for dataset balancing and scikit-learn for splitting and evaluating the data.

4.2. Data split for training and testing

The dataset was split into training and testing sets using 80 % of the data for training and 20 % for testing. This training and testing split is applied in both cases, such as after balancing the complete dataset using augmentation techniques and before applying augmentation techniques. To evaluate the real-world scenario performance of a balanced dataset, the models are trained on a balanced augmented dataset and then tested with an actual imbalanced dataset.

4.3. Performance evaluation metrics

The models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices. These metrics were crucial for assessing the performance of the models on both classes of solar panel images, ensuring that the models could effectively detect dusty panels in real-world scenarios.

4.3.1. Accuracy

Accuracy is the most common evaluation metric, measuring the overall percentage of correct predictions made by the model (Vujović et al., 2021). Accuracy is the ratio of correct predictions (both true positives and true negatives) to the total number of samples. In the context of solar panel dust detection, accuracy helps determine how often the model correctly classifies both classes of panels. However, in imbalanced datasets, high accuracy can be misleading as it might heavily favor the majority class (clean panels). Thus, other metrics must be used in tandem to gain a more nuanced understanding. It is calculated as:

Table 6

Computational environment and resources used in proposed research.

Computing Resources	Details
CPU	Intel(R) Core(TM) i7-13700 (24 CPUs) 2.1 GHz
GPU	NVIDIA GeForce RTX 4060
RAM	16 GB
System	64-bit operating system, x64-based processor
Language	Python (3.9.19)
Framework	Keras(2.7.0) with TensorFlow(2.7.0) backend
Software	VisualStudio

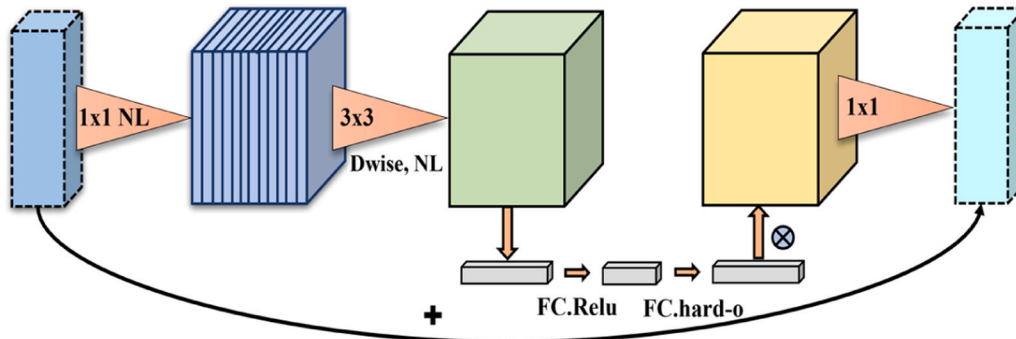


Fig. 10. MobileNetV3 Architecture: Illustrating the use of 1×1 and depthwise separable 3×3 convolutions (Dwise), nonlinear (NL) activation functions, and lightweight bottleneck structures with squeeze-and-excitation (SE) modules for efficient feature extraction. The skip connections and fully connected layers (FC) facilitate enhanced information flow and improved model performance for solar panel dust classification.

$$\text{Accuracy} = \frac{TP + FN}{TP + TN + FP + FN} \quad (3)$$

4.3.2. Precision

Precision measures the proportion of positive predictions that are correct (Yao and Shepperd, 2021). In this research, high precision ensures that when the model predicts a panel as dirty, it actually contains dirt, thereby helping to avoid unnecessary cleaning of clean panels. Precision is particularly important when the cost of cleaning a panel is high. It is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

4.3.3. Recall

Recall, or Sensitivity, evaluates the model's ability to identify all relevant instances (Cheema and Urner, 2023). In solar panel dust detection, high recall ensures that most dusty panels are identified, thereby reducing the risk of overlooking those that require cleaning. It is crucial to minimize the potential impact of missed predictions on solar panel efficiency. It is given by:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

4.3.4. F1-score

The F1-Score combines precision and recall into a single metric, providing a balance between the two. The F1-Score balances the trade-off between precision and recall, providing a single metric that considers both false positives and false negatives (Diallo et al., 2024). This is especially useful in imbalanced datasets where one class is underrepresented. The F1-Score is the harmonic mean of precision and recall and is defined as:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

4.4. Confusion matrix

The confusion matrix provides a detailed breakdown of the model's predictions, showing the counts of true positives, true negatives, false positives, and false negatives (Amin and Mahmoud, 2022). In addition to the overall accuracy, precision, and recall, confusion matrices were used to identify areas where the models performed well and areas where improvements were needed. It is essential for visualizing how well the model distinguishes between clean and dusty panels and understanding where the model is making errors. These metrics are particularly useful in our research on solar panel dust detection as they ensure the model is both accurate and reliable in identifying contaminated panels, which is critical for efficient maintenance and maximizing energy efficiency. By using these evaluation metrics, it can assess the trade-offs between correctly identifying dirty panels and avoiding false alarms, which is especially important in real-world applications where resources for cleaning panels are limited.

4.5. Cross validation and robustness evaluation

To address the risk of overfitting on a relatively small and imbalanced dataset, and in accordance with established best practices for robust model evaluation (Aliferis and Simon, 2024), we adopted a rigorous stratified 10-fold cross-validation (CV) protocol for all experiments. In each fold, 90 % of the data were used for model training including the application of SMOTE or Stable Diffusion balancing only on the training fold and 10 % for independent testing. This ensures that synthetic images and augmented samples do not leak into the test set, providing an unbiased estimate of model generalization. All performance metrics (accuracy, precision, recall, and F1-score) are reported as

the mean $\pm$ standard deviation over the 10 folds, offering model-agnostic confidence intervals and variance estimates. Such evaluation protocols are strongly recommended for small, imbalanced image datasets and directly address the risk of overfitting associated with high-capacity CNNs and limited data (Islam et al., 2024).

4.6. Cohen's Kappa (κ) and Matthews correlation coefficient (MCC)

Cohen's kappa κ and the Matthews Correlation Coefficient (MCC) are two robust evaluation metrics that are particularly valuable for assessing classification performance in the presence of class imbalance (Yilmaz and Demirhan, 2023). In this research, we include both Cohen's kappa and MCC alongside conventional metrics to provide a statistically rigorous evaluation of our deep-learning models for solar panel dust detection. By doing so, we aim to assess not only overall performance but also the true discriminative power and robustness of our models under class imbalance.

4.6.1. Cohen's Kappa (κ)

Cohen's kappa κ measures the agreement between predicted and true class labels while correcting for the agreement expected by random chance, thereby providing a more nuanced assessment than accuracy alone. The kappa statistic is calculated as:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (7)$$

Where p_o is the observed accuracy (the proportion of samples where predicted and actual labels match) and p_e is the probability of random agreement, estimated from the distribution of the labels. Kappa values range from -1 (complete disagreement) to 1 (perfect agreement), with 0 indicating agreement equivalent to chance. Prior studies (Yilmaz and Demirhan, 2023) have established kappa as a standard measure for evaluating classifiers on imbalanced datasets, as it provides insight into whether high accuracy is genuinely meaningful or simply due to class distribution.

4.6.2. Matthews Correlation Coefficient (MCC)

The Matthews Correlation Coefficient (MCC) is another comprehensive metric defined as:

$$\text{MCC} = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (8)$$

Where TP, TN, FP, and FN denote the counts of true positives, true negatives, false positives, and false negatives, respectively. MCC produces a value between -1 (total disagreement) and 1 (perfect prediction), with 0 indicating random prediction. Unlike accuracy and F1-score, MCC takes into account all four confusion matrix categories and is widely recognized as one of the most reliable measures for binary classification, especially when class sizes are unequal (Chicco et al., 2021).

4.7. Adaptations in this study

While CNNs, SMOTE, and diffusion models are well established in the literature, this study applied them with several specific adaptations.

1. SMOTE was employed in the feature vector space as a classical oversampling baseline, allowing us to benchmark the improvement gained from image-level generative augmentation. Although this omits spatial correlations, it provides a useful contrast to Stable Diffusion.
2. Stable Diffusion was customized for PV dust detection by using prompt-driven generation and post-processing steps to crop and filter images, ensuring that only realistic panel regions were included and that no overlap occurred with validation/test sets.

3. The CNNs (VGG-16, ResNet50, MobileNetV3) were not altered architecturally, but were systematically evaluated under a stratified 10-fold cross-validation framework with imbalance-sensitive metrics (F1-score, Cohen's κ , and MCC), and confusion-matrix-based reporting of TP, TN, FP, and FN values.

These adaptations distinguish our implementation from standard tutorials and focus the methodology on the unique requirements of small, imbalanced PV datasets.

5. Results

This section presents the experimental results of the proposed models on both imbalanced and balanced datasets (SMOTE and Stable Diffusion). Model performance is evaluated using standard metrics, including accuracy, precision, recall, and F1-score, based on a stratified 10-fold cross-validation protocol. The results focus on quantitative comparisons among different data balancing strategies and model architectures, providing an objective assessment of the effectiveness of each approach for solar panel dust detection.

5.1. Performance of CNN models on imbalanced dataset

The results in Fig. 11 illustrate that, in all the metrics considered, VGG-16 (85.62 %) is better than ResNet50 (76.53 %) and MobileNetV3 (80.05 %). On the imbalanced dataset, the accuracy results for dusty panels are measured as VGG-16 to be 86 %, then ResNet50 and MobileNetV3, which performed lower accuracies compared to the minority class with a 96 % recall rate on the minority class, showing that VGG-16 is good at detecting the minority class, especially for majority panels, as shown in Fig. 11. The confusion matrix components (TN, TP, FN, FP) for all models under imbalanced, Stable Diffusion, and SMOTE conditions are summarized in Table 7. Under the imbalanced setting, VGG-16 achieved TN = 97 and TP = 49 with misclassifications FN = 19 and FP = 4. Both augmentation strategies improved performance: Stable Diffusion reduced false negatives for ResNet50 from FN = 34 (imbalanced) to FN = 2, while SMOTE enabled VGG-16 to achieve TP = 87 with FN = 13, indicating stronger minority-class detection, as shown in Table 7.

Furthermore, using VGG-16 shows that precision is highest (92 % for panels with dirt), indicating a lower number of false positives. The F1-scores demonstrate that VGG-16 achieves a reasonable balance

Table 7

Confusion matrix components (TN, TP, FN, FP) for each model under imbalanced, Stable Diffusion, and SMOTE augmentation conditions.

Augmentation Type	Model	TN	TP	FN	FP
Imbalanced	VGG-16	97	49	19	4
	ResNet50	95	34	34	6
	MobileNetV3	94	42	26	7
Stable Diffusion	VGG-16	100	76	24	1
	ResNet50	101	98	2	0
	MobileNetV3	95	69	31	6
SMOTE	VGG-16	100	87	13	1
	ResNet50	97	62	38	4
	MobileNetV3	96	70	30	5

between precision and recall. This is confirmed by the validation accuracy plot shown in Fig. 12, as all the models perform worse than the VGG-16. Although MobileNetV3 is lightweight, due to having less capacity than other models, it failed to detect dirty panels more. Our results indicate that the deeper architecture of VGG-16 is more suitable for learning discriminative image features for dust detection on solar panels and handling imbalanced data. For that sake, VGG-16 is the best-suited model for this task in terms of accuracy and generalization. The evaluation of the confusion matrix indicates that VGG-16 produces the most accurate and balanced predictions, and therefore, it is the best model for detecting dust on solar panels. Table 8 shows the performance of CNN models evaluated on the imbalanced dataset.

5.2. Performance of CNN models on balanced dataset

5.2.1. Balanced with SMOTE

The results in the balanced dataset with SMOTE indicate that all models (VGG-16, ResNet50, and MobileNetV3) have performed significantly better across all metrics compared to the imbalanced dataset, with VGG-16 achieving the highest performance across all metrics shown in Table 8. The classification histogram in Fig. 13 indicates that the VGG-16 has the highest precision of 93.5 % compared to ResNet50 (79.82 %) and MobileNetV3 (82.56 %), which is defined as the proportion of correctly predicted panels with dirt. On the other hand, recall is 99 %, which indicates the model's capability to identify dirty panels among clean and dusty solar panels, and suggests good calibration for detecting both classes. VGG-16 achieves the highest F1-Score of 95 %, confirms that it's performing optimally in comparison to the other

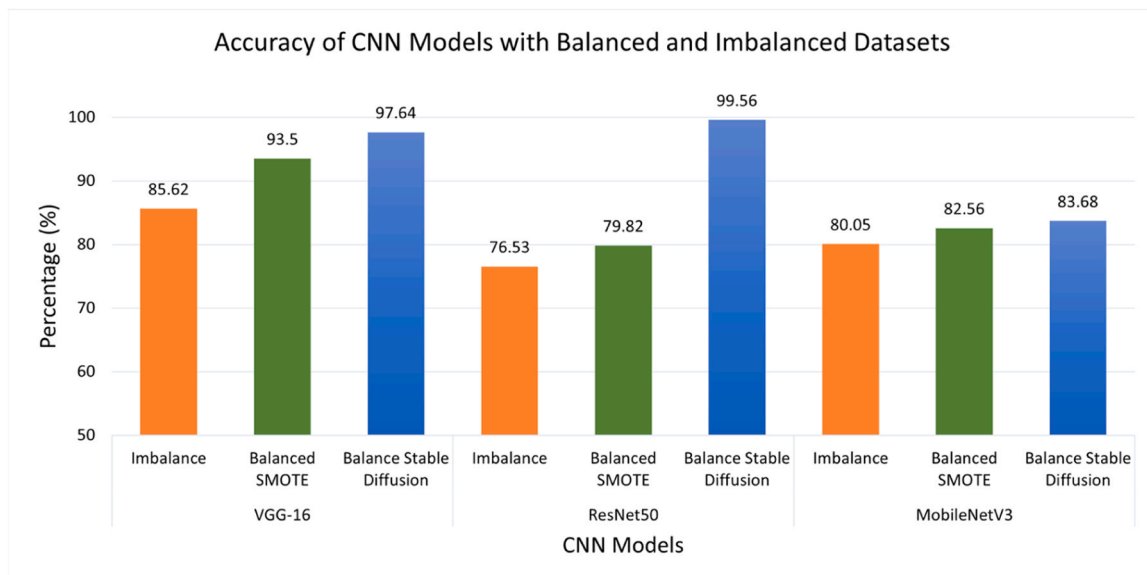


Fig. 11. Accuracy performance of VGG-16, ResNet50, and MobileNetV3 on imbalanced and balanced datasets.

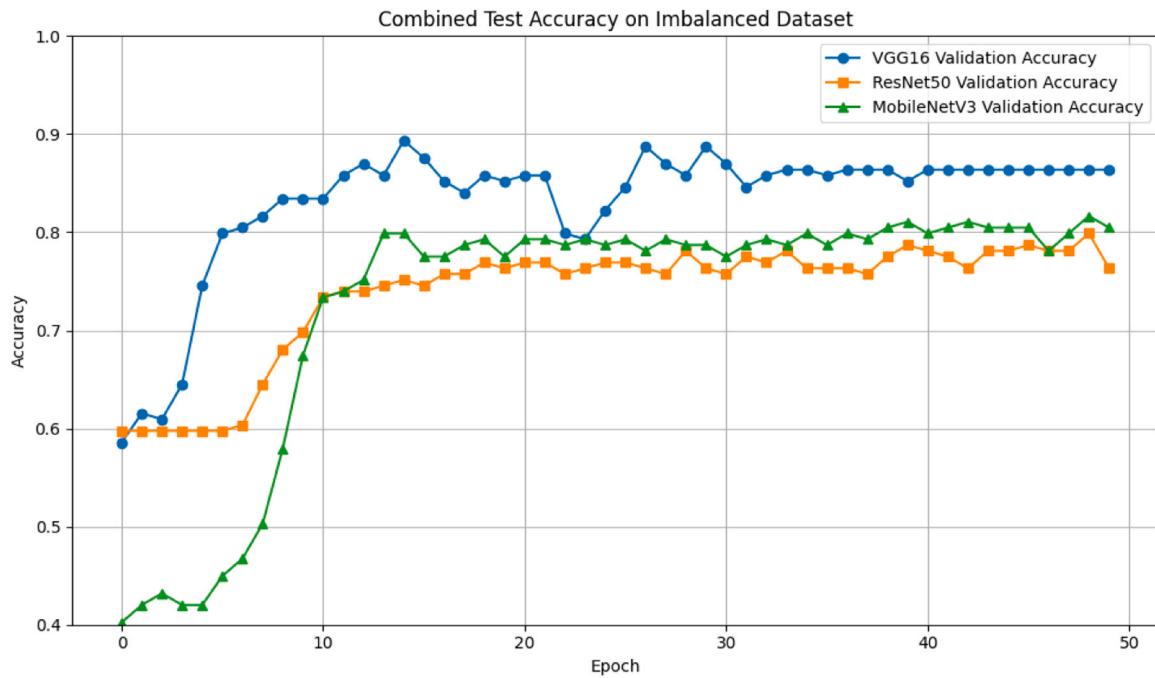


Fig. 12. Combined accuracy results of models on imbalanced dataset.

Table 8

Classification report of VGG-16, ResNet50, and MobileNetV3 under different augmentation settings (balanced and imbalanced).

Augmentation	Class	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Imbalanced	Dirty	VGG-16	85.62	92	72	81
		ResNet50	76.53	85	50	63
		MobileNetV3	80.05	86	62	72
SMOTE	Clean	VGG-16	85.62	84	96	81
		ResNet50	76.53	74	94	83
		MobileNetV3	80.05	78	93	85
	Dirty	VGG-16	93.50	99	90	94
		ResNet50	79.82	94	62	75
		MobileNetV3	82.56	93	70	80
	Clean	VGG-16	93.50	91	99	95
		ResNet50	79.82	72	96	82
		MobileNetV3	82.56	76	95	85
Stable Diffusion	Dirty	VGG-16	97.64	99	94	96
		ResNet50	98.87	99	99	99
		MobileNetV3	83.68	93	81	87
	Clean	VGG-16	97.64	94	99	97
		ResNet50	98.87	99	99	99
		MobileNetV3	83.68	77	95	85

models, and completely balances the precision and recall as shown in the histogram in Fig. 13. The confusion matrices shown in Table 7 also support the fact that VGG-16 was the best-performing model among all the models tested. In VGG-16, it is evaluated that the matrix has very few false positives and false negatives, which indicates that it should be a reliable and precise model in terms of clean and dusty classifications. Thus, VGG-16 is more robust for cleaning dirty panels, with fewer errors. The combined accuracy graph in Fig. 14 shows a very good accuracy curve, indicating that VGG-16 converged much more quickly than ResNet50 and MobileNetV3, reaching a high accuracy of about 95 %. Whereas MobileNetV3 initially performed well on an imbalanced dataset, achieving an accuracy of 80.05 %, on a balanced dataset with SMOTE and Stable Diffusion, the model's performance is moderate, with accuracies of 85.56 % and 83.68 %, respectively. The model has demonstrated efficient learning and the ability to generalize through this rapid convergence. MobileNetV3, meanwhile, showed the slowest convergence, failing to achieve good accuracy.

5.2.2. Balanced with stable diffusion

For the balanced dataset after applying Stable Diffusion, the performance evaluation results indicate a noticeable improvement in model performance across all metrics shown in Table 8. Among the three models (VGG-16, ResNet50, MobileNetV3), ResNet50 performs best with an accuracy of 98.87 % in terms of panel detection, especially in panels with dirt, as compared to the accuracies of VGG-16 (97.64 %) and MobileNetV3 (83.68 %). Classification histograms shown in Figs. 15 and 16 indicate that classification accuracy, precision, recall, and F1-score of dirty panels are significantly higher for ResNet50 in comparison with VGG-16 and MobileNetV3. The confusion matrix shown in Table 7 also clarifies that ResNet50 has a very good ability to identify panels with dirt (98 true positives) but relatively few false negatives and false positives. Therefore, ResNet50 outperforms other models for this task, especially in dirty panel classification. The results indicate that using Stable Diffusion to balance the dataset has helped increase the model's accuracy across all evaluation metrics.

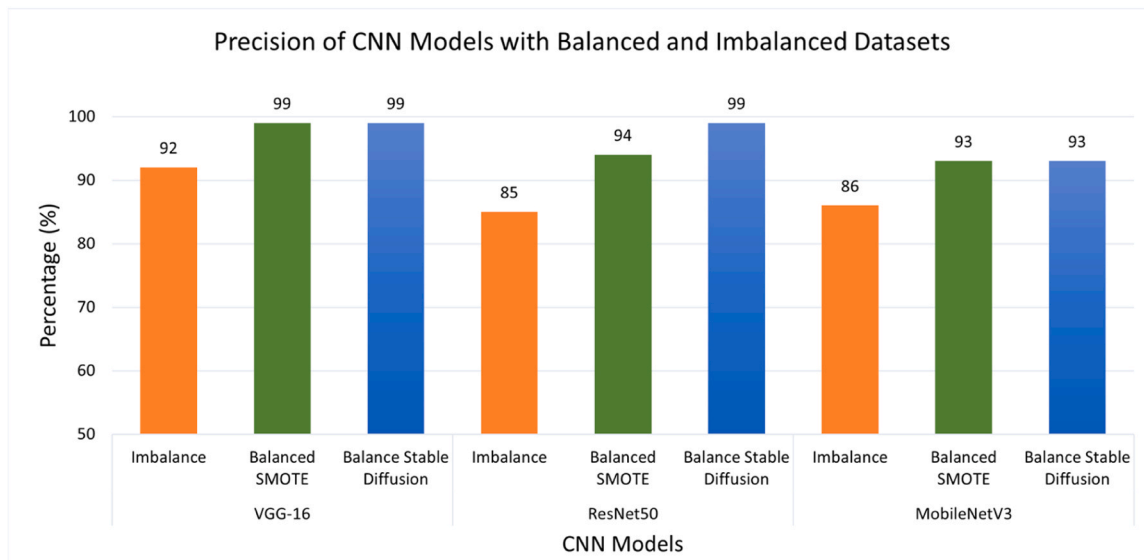


Fig. 13. Precision performance of VGG-16, ResNet50, and MobileNetV3 on balanced and imbalanced datasets.

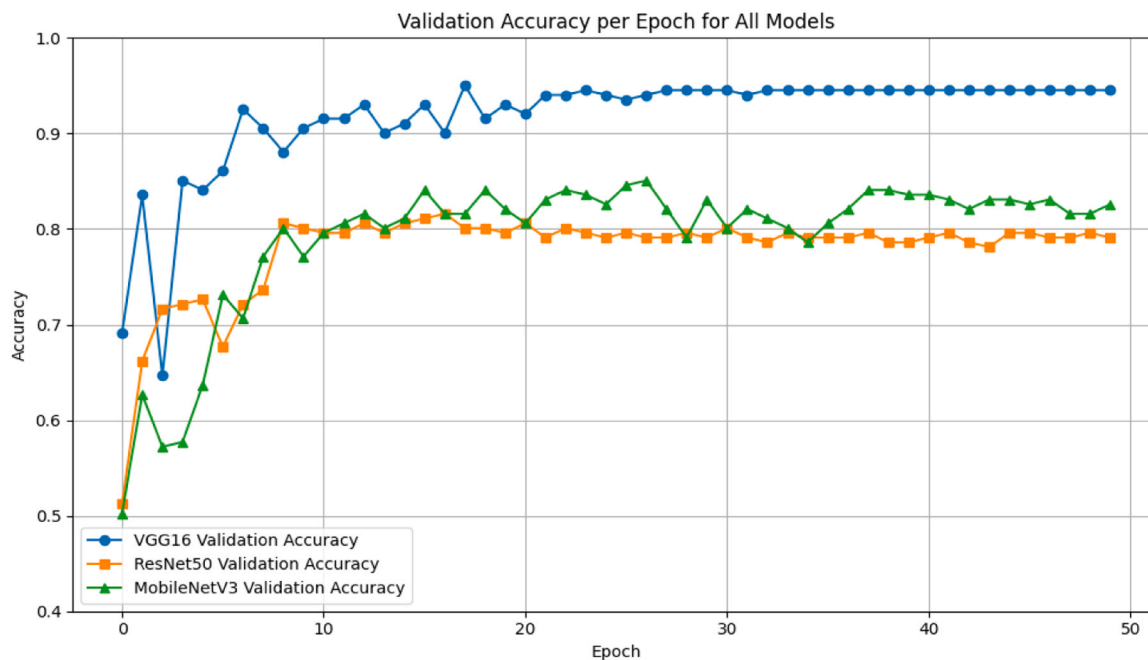


Fig. 14. Model's combined accuracy results with balanced dataset using SMOTE.

5.3. Real-world experiment: training on balanced set and testing on imbalanced data

To evaluate the performance of the proposed augmentation strategies under deployment-like conditions, we conducted experiments where models were trained on balanced datasets (using SMOTE or Stable Diffusion) but tested on the original imbalanced test sets. This setup simulates a real-world scenario in which PV monitoring systems must operate under naturally skewed data distributions. The goal is to assess whether the benefits of augmentation during training generalize to imbalanced test conditions, thereby providing stronger evidence of model reliability in practical applications. These findings confirm that models trained with Stable Diffusion augmentation remain robust when deployed on naturally imbalanced test sets.

5.3.1. Training set balanced with SMOTE

Balancing the training set using SMOTE significantly improved model performance on the imbalanced test set. As shown in the combined accuracy curves in Fig. 17, VGG-16, ResNet50, and MobileNetV3 achieved validation accuracies above 94 %, indicating generalization to the imbalanced test set. The confusion matrices reveal that all models maintained high precision and recall for both classes, with results shown in Table 9. VGG-16 exhibited the best performance, achieving 98 % accuracy, whereas ResNet50 also outperformed, achieving 96 % accuracy, and MobileNetV3 achieved 94 % accuracy. These results classify nearly all samples with minimal misclassification. ResNet50 and MobileNetV3 also delivered strong results, with minor confusion between clean and dirty classes. Overall, SMOTE balancing effectively mitigated the effects of class imbalance, enhancing model reliability in practical scenarios.

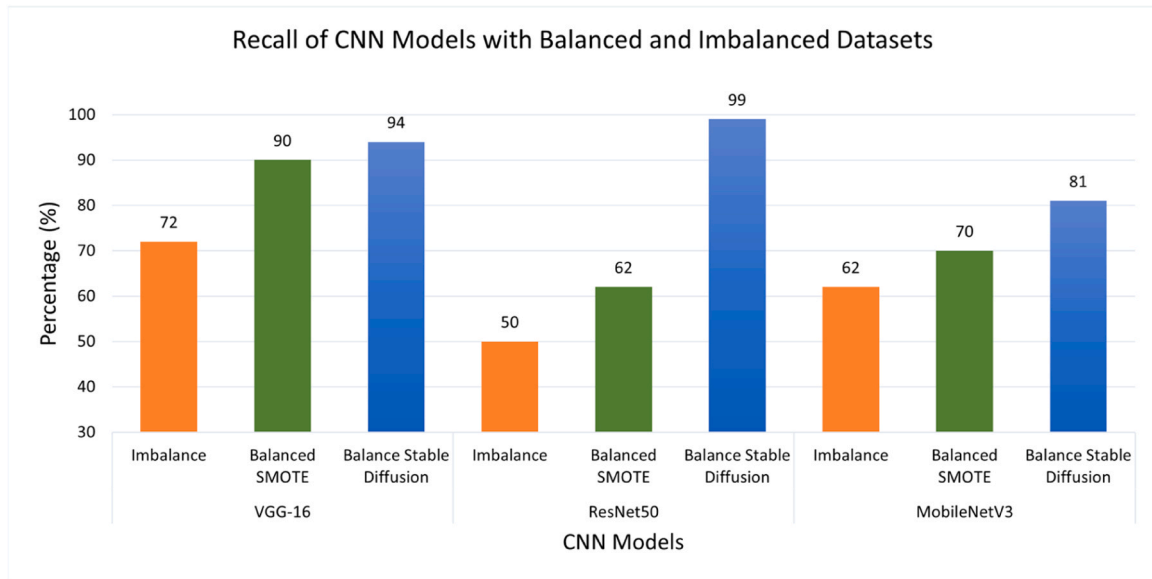


Fig. 15. Recall performance of VGG-16, ResNet50, and MobileNetV3 on balanced and imbalanced datasets.

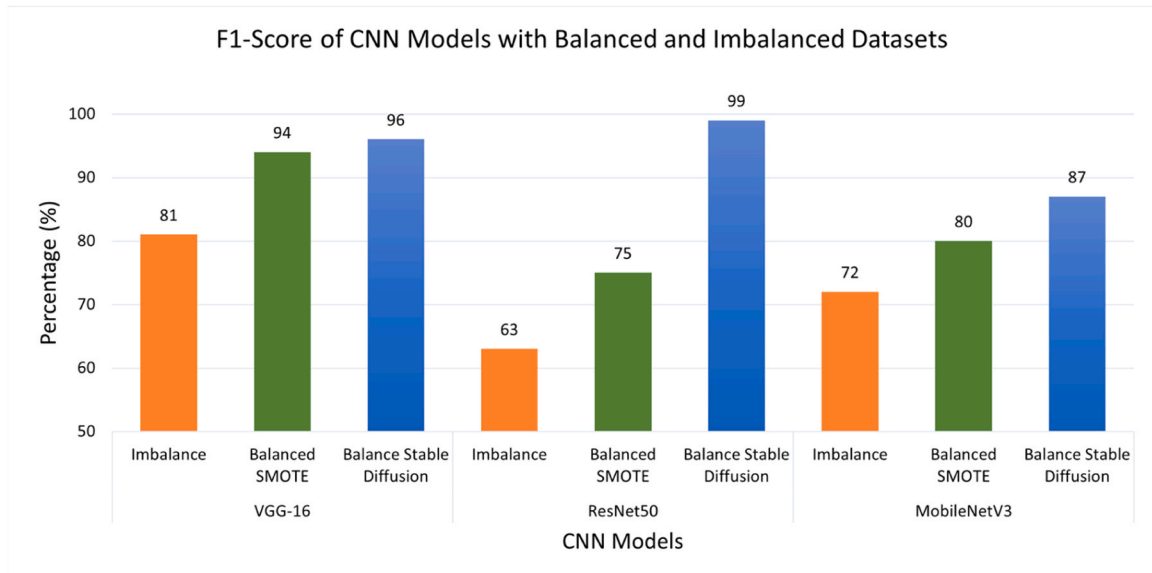


Fig. 16. F1-Score of VGG-16, ResNet50, and MobileNetV3 on balanced and imbalanced datasets.

5.3.2. Training set balanced with stable diffusion

Balancing the training set using augmentation with Stable Diffusion-generated images led to strong classification performance across all models. As shown in Fig. 18, validation accuracy reached 96 % for VGG-16, 93 % for ResNet50, and 96 % for MobileNetV3. The confusion matrices (see Table 9) reveal VGG-16 correctly classified 97 clean and 66 dirty samples, while MobileNetV3 and ResNet50 achieved similar results with high precision (up to 97 %) and recall (up to 98 %) for both classes, as discussed in Fig. 19. The confusion matrix (see Table 9) components (TN, TP, FN, FP) for models trained on balanced datasets using Stable Diffusion and SMOTE augmentation are presented in Table 9. Overall, misclassification rates are low across both strategies ($FN \leq 5$ and $FP \leq 8$). In this deployment-like evaluation, VGG-16 achieved the best overall performance (reported accuracy 98 %), while ResNet50 and MobileNetV3 achieved approximately 96 % and 94 % accuracy, respectively. These improvements confirm that augmented data enriches model learning, yielding superior generalization on the

imbalanced test set.

5.4. Cross validation results and overfitting risk

Table 10 summarizes the 10-fold cross-validation results for all models and data balancing strategies. Each metric is reported as mean $\pm$ standard deviation across the 10 folds. The use of cross-validation provides a more conservative and reliable estimate of model performance compared to a single train-test split, as recommended in recent machine learning literature (Islam et al., 2024). Table 10 presents the average accuracy, precision, recall, and F1-score, along with their respective standard deviations, for each model and balancing strategy. The results demonstrate that balancing the dataset with SMOTE or Stable Diffusion substantially improves recall and F1-score, particularly for detecting the minority (dusty) class. Notably, the best overall performance is achieved by ResNet50 and VGG-16 on the Stable Diffusion balanced dataset, with mean accuracy and F1-score exceeding 95 %. The low standard

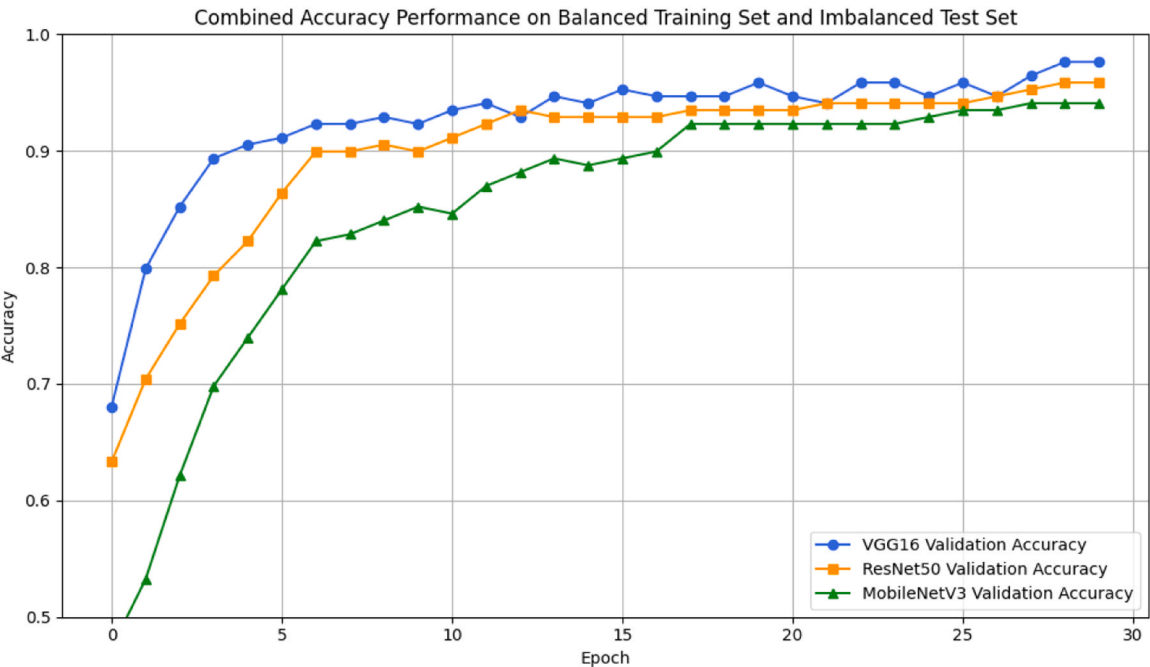


Fig. 17. Model’s combined accuracy results on balanced training set with SMOTE.

Table 9
Confusion matrix components (TN, TP, FN, FP) for models trained on balanced datasets (real-world scenario) using Stable Diffusion and SMOTE augmentation.

Augmentation	Model	TN	TP	FN	FP
Stable Diffusion	VGG-16	100	65	3	1
Stable Diffusion	ResNet50	97	65	3	4
Stable Diffusion	MobileNetV3	94	65	3	7
SMOTE	VGG-16	97	66	2	4
SMOTE	ResNet50	93	64	4	8
SMOTE	MobileNetV3	99	63	5	2

deviations across all folds further confirm the robustness and consistency of these results. These trends not only validate the effectiveness of our approach but also align with findings from previous studies on deep learning for imbalanced image classification (Haque et al., 2025).

5.5. Statistical significance and variance analysis

To validate the reliability of the observed improvements, we performed Wilcoxon Signed Rank tests across the 10-fold cross-validation results for Accuracy and F1-score. As shown in Table 11, Stable Diffusion consistently outperformed both the imbalanced baseline and SMOTE across the majority of cases. For VGG-16 and ResNet50, the

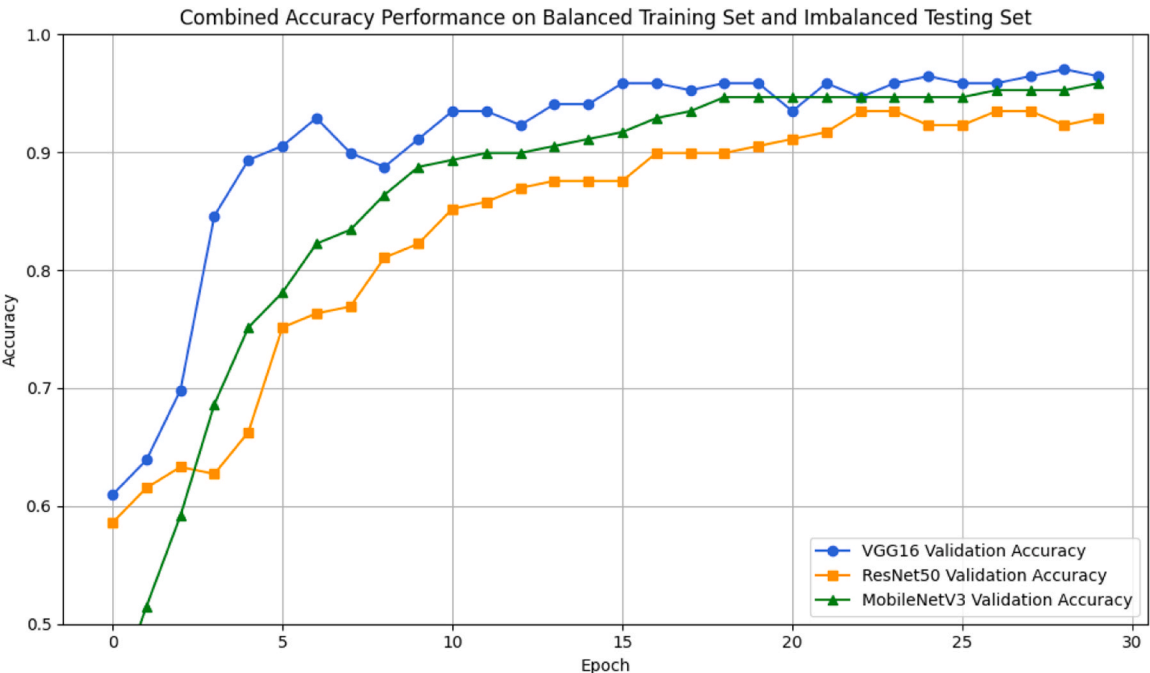


Fig. 18. Model’s combined accuracy results on balanced training set with Stable Diffusion.

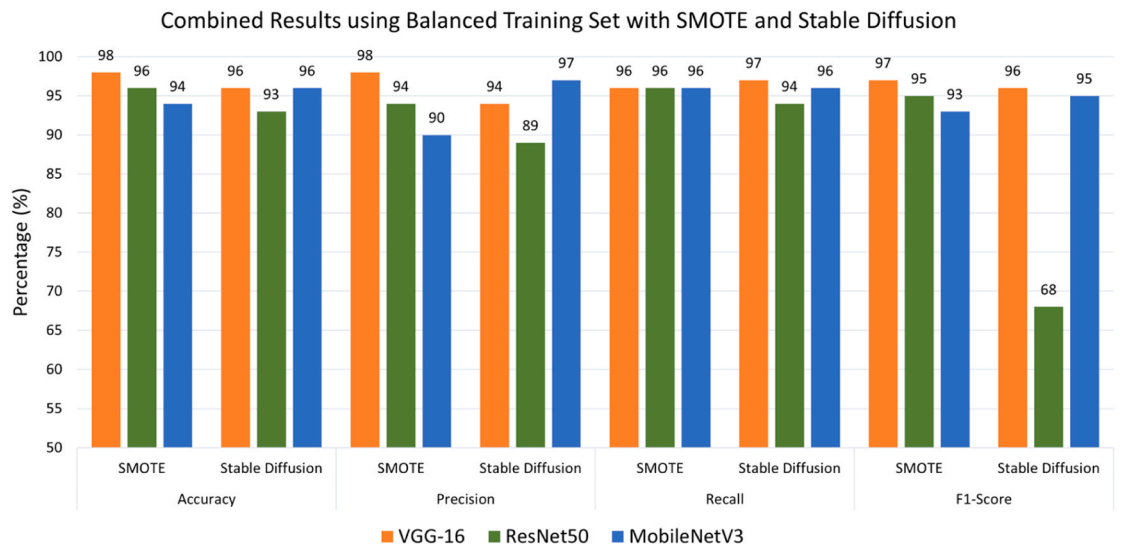


Fig. 19. Combined classification report of models on balanced training and imbalanced testing set.

Table 10

Performance of VGG-16, ResNet50, and MobileNetV3 using 10-fold cross-validation (Mean $\pm$ SD).

Technique	Model	Accuracy	Precision	Recall	F1-Score
Imbalance	VGG-16	83.72 ($\pm$ 0.043)	90.65 ($\pm$ 0.028)	71.14 ($\pm$ 0.098)	80.90 ($\pm$ 0.079)
	ResNet50	74.90 ($\pm$ 0.048)	83.23 ($\pm$ 0.097)	52.02 ($\pm$ 0.112)	63.16 ($\pm$ 0.092)
	MobileNetV3	79.12 ($\pm$ 0.049)	86.45 ($\pm$ 0.066)	64.03 ($\pm$ 0.084)	72.03 ($\pm$ 0.057)
Balance w/ SMOTE	VGG-16	91.56 ($\pm$ 0.024)	97.32 ($\pm$ 0.009)	88.15 ($\pm$ 0.030)	92.34 ($\pm$ 0.018)
	ResNet50	77.89 ($\pm$ 0.050)	90.75 ($\pm$ 0.073)	65.01 ($\pm$ 0.100)	76.68 ($\pm$ 0.052)
	MobileNetV3	81.97 ($\pm$ 0.036)	92.91 ($\pm$ 0.031)	71.09 ($\pm$ 0.063)	80.98 ($\pm$ 0.033)
Balance w/ Stable Diffusion	VGG-16	95.55 ($\pm$ 0.017)	98.88 ($\pm$ 0.011)	91.80 ($\pm$ 0.020)	95.15 ($\pm$ 0.016)
	ResNet50	97.21 ($\pm$ 0.014)	98.79 ($\pm$ 0.013)	97.13 ($\pm$ 0.009)	97.94 ($\pm$ 0.011)
	MobileNetV3	82.18 ($\pm$ 0.040)	91.70 ($\pm$ 0.040)	82.61 ($\pm$ 0.045)	86.76 ($\pm$ 0.026)

gains of Stable Diffusion over both Imbalance and SMOTE were statistically significant ($p < 0.05$), with particularly strong evidence in ResNet50 where the F1-score reached $p < 0.01$. MobileNetV3 also demonstrated improvements with Stable Diffusion, although the differences were smaller and, in some cases, only marginally significant. These results confirm that the enhancements introduced by Stable Diffusion are not due to random variation but represent meaningful performance gains, especially for deeper architectures such as ResNet50.

5.6. Kappa and MCC analysis

Cohen's kappa (κ) and Matthews Correlation Coefficient (MCC)

Table 11

Wilcoxon Signed Rank Test Results (10-fold CV) for F1-Score Across Models and Augmentation Methods.

Model	Comparison	Statistic	p-value	Significant
VGG-16	Imbalance vs SMOTE	15.0	0.0312	Yes
	Imbalance vs Stable Diffusion	15.0	0.0039	Yes
	SMOTE vs Stable Diffusion	14.0	0.0469	Yes
ResNet50	Imbalance vs SMOTE	13.0	0.0625	Borderline
	Imbalance vs Stable Diffusion	15.0	0.0039	Yes
	SMOTE vs Stable Diffusion	15.0	0.0039	Yes
MobileNetV3	Imbalance vs SMOTE	12.0	0.1250	No
	Imbalance vs Stable Diffusion	14.0	0.0469	Yes
	SMOTE vs Stable Diffusion	13.0	0.0625	Borderline

provide a robust assessment of model performance by quantifying both agreement with ground truth and the balanced ability to predict minority and majority classes. In this study, both metrics clearly reflect the trends observed across all models and balancing strategies. Table 12 presents κ and MCC for all models and balancing techniques, evaluated using 10-fold cross-validation. Both metrics offer a balanced and chance-corrected assessment of model performance, making them especially informative for imbalanced classification tasks. Under the imbalanced setting, all models achieve only moderate agreement and balanced prediction (e.g., $\kappa = 0.61$, MCC = 0.65 for VGG-16), with ResNet50 showing the lowest scores. The use of SMOTE leads to significant improvements, particularly for VGG-16 and MobileNetV3, with κ and MCC

Table 12

κ and MCC for models Using 10-fold Cross-Validation (Mean $\pm$ SD).

Technique	Model	κ ($\pm$ SD)	MCC ($\pm$ SD)
Imbalance	VGG-16	0.61 ($\pm$ 0.08)	0.65 ($\pm$ 0.07)
	ResNet50	0.37 ($\pm$ 0.10)	0.43 ($\pm$ 0.09)
	MobileNetV3	0.50 ($\pm$ 0.09)	0.56 ($\pm$ 0.08)
Balance w/ SMOTE	VGG-16	0.81 ($\pm$ 0.04)	0.84 ($\pm$ 0.03)
	ResNet50	0.41 ($\pm$ 0.09)	0.53 ($\pm$ 0.08)
	MobileNetV3	0.59 ($\pm$ 0.08)	0.68 ($\pm$ 0.07)
Balance w/ Stable Diffusion	VGG-16	0.91 ($\pm$ 0.02)	0.93 ($\pm$ 0.02)
	ResNet50	0.95 ($\pm$ 0.01)	0.96 ($\pm$ 0.01)
	MobileNetV3	0.67 ($\pm$ 0.06)	0.74 ($\pm$ 0.05)

increasing to 0.81 and 0.84, respectively.

The greatest gains are seen with Stable Diffusion, where both VGG-16 and ResNet50 achieve κ and MCC above 0.90, indicating almost perfect agreement and highly robust, balanced classification. These results demonstrate that the improvements seen in accuracy and F1-score are not merely artifacts of class distribution but reflect genuine advances in both predictive power and agreement with true labels, as validated by the most widely recommended statistical metrics for imbalanced data. While overall accuracy remained high across all models, κ and MCC revealed clearer differences between augmentation methods. For example, Stable Diffusion improved κ and MCC more substantially than SMOTE, reflecting that its gains extended beyond accuracy to better agreement and balanced performance. Precision and recall showed expected trade-offs: in some cases recall improved at the expense of a small drop in precision, highlighting that models became more sensitive to detecting dusty panels while tolerating a few additional false alarms. This behavior is advantageous in PV maintenance, where false negatives (undetected dusty panels) are more costly than false positives.

6. Discussion

This section interprets and contextualizes the experimental results, discusses key trends and comparative outcomes, evaluates the strengths and limitations of the proposed approach, and considers practical implications for solar panel dust detection and future research directions.

6.1. Evaluation of synthetic images

To evaluate the quality and reliability of synthetic images generated through Stable Diffusion, we employed a two-stage evaluation framework that comprised manual image assessment and embedding similarity tests in the feature space. In the first stage, a total of 300 synthetic images were screened using the evaluation rubric (see Table 4) and a two-step filtering process involving manual inspection and perceptual hashing. This process eliminated 138 images due to unrealistic textures, geometric distortions, or redundancy, while retaining 162 high-quality images. These retained samples were selected based on their plausibility compared to real-world PV panel images, with emphasis on dust pattern realism, panel geometry integrity, and lighting consistency. In the second stage, quantitative validation was performed using FID, KID, and pHash to compare the distribution of synthetic images with real PV panel images. As summarized in Table 5, the results showed progressive improvements: FID decreased from 107.65 (baseline) to 63.86 after calibration and filtering, while KID reduced from 45.55×10^{-3} to 21.99×10^{-3} . The pHash duplicate rate remained 0 %, confirming diversity among generated samples.

6.2. Evaluation of data augmentation techniques

The results from the comparison between SMOTE and Stable Diffusion on the balanced dataset demonstrate the effectiveness of both techniques in enhancing model performance for dust detection on solar panels. From these results, it is observed that applying both data augmentation methods yields more significant and reliable evaluation metric results. It is clear from the histogram shown in Fig. 16 that Stable Diffusion outperforms SMOTE in almost all cases, with results of 98.87 % accuracy for ResNet50 and 97.64 % accuracy for VGG-16 in detecting clean and dusty panels. Comparing the performance of models against SMOTE, Stable Diffusion results in higher and more accurate predictions of dust accumulation on solar panels. As shown in Fig. 16, SMOTE achieves comparatively better accuracies for the VGG-16 model, with 93.5 % accuracy. ResNet50 achieves a slightly lower prediction accuracy with SMOTE, at 79.82 %, whereas VGG-16 performs well in this case, with 93.5 % accuracy. Furthermore, the results indicate that, with Stable Diffusion augmentation techniques, ResNet50 yields very significant results, achieving an accuracy of 98.87 %. VGG-16 also

performs well, with an accuracy of 97.64 %. It is observed that Stable Diffusion's key advantage comes from its ability to produce realistic, high-quality synthetic images that are consistent with real-world conditions. Thus, it is better equipped to spot patterns in dusty solar panels. At the same time, SMOTE creates synthetic examples through interpolation, but these images may not perfectly replicate the intricacies of real-world dust patterns. The results suggest that Stable Diffusion augmentation produces more accurate and reliable model predictions with fewer false positives and false negatives and thus constitutes a more reliable data augmentation technique for the task of dust detection. Finally, it is concluded that Stable Diffusion is a more suitable and reliable data augmentation technique for CNN models used in predicting dust accumulation on solar panels. The advantage of SMOTE is that it provides higher accuracy, overgeneralization, and overall performance advantages.

6.3. Evaluation of CNN models with balanced and imbalanced datasets

The models' performance based on evaluation metrics of accuracy, precision, recall, and F1-score shows that CNN models are quite effective in dust detection on solar panels, especially in the case of the imbalanced dataset. From the results shown in Figs. 11, 13, 15, and 16, it is observed that models trained on the imbalanced dataset achieve lesser accuracy and performance, especially in detecting the dirty solar panel, with low precision and recall values of the "dirty" class. Due to the imbalance, the dataset model results in biased predicted labels that favor the majority class, thereby limiting its effectiveness in identifying dirty panels. However, the models perform quite well when data augmentation techniques, e.g., SMOTE and Stable Diffusion, are applied to balance the dataset. From the results for VGG-16 with the imbalanced dataset, it is observed that the prediction accuracy is 85.62 %. However, after balancing the dataset with SMOTE and Stable Diffusion, accuracy results of 93.5 % and 97.64 %, respectively, are achieved. Similarly, with ResNet50, it initially achieved an accuracy of 76.53 % on the imbalanced dataset. However, after balancing the dataset with SMOTE and Stable Diffusion, it achieved increased accuracies of 79.82 % and 98.87 %, respectively. These results demonstrate that data balancing is necessary for accurate dust prediction in solar panels, yielding significant performance improvements. These results demonstrate that proper balancing of the dataset is vital for reliable prediction and cannot be achieved without data augmentation strategies, such as SMOTE and Stable Diffusion, to overcome the problem of imbalanced data in solar panel dust detection systems.

6.4. Evaluation of CNN models with balanced training set using SMOTE and stable diffusion

Both SMOTE and Stable Diffusion-based augmentation were employed to balance the training set, and their performance was evaluated on the original imbalanced test set, yielding substantial improvements in model performance, as shown in Table 9 and 19. When balanced with SMOTE, models achieved high validation accuracies up to 98 % for VGG-16 and maintained strong recall and precision (e.g., VGG-16: 98 % macro-averaged F1-score), demonstrating SMOTE's effectiveness in generating diverse synthetic samples at the feature level. However, Stable Diffusion-based augmentation further enhanced robustness, with VGG-16 and MobileNetV3 achieving 96 % validation accuracy and exhibiting fewer misclassifications, as shown in the confusion matrices in Table 9. This can be attributed to the ability of image-based augmentation to introduce more realistic and varied samples, enabling the models to better capture subtle features and class boundaries, especially for the minority class. These improvements align with (Sagers et al.,), which emphasizes that advanced augmentation methods can better address class imbalance and promote generalization. Overall, both strategies significantly reduce bias, but Stable Diffusion offers an added advantage by creating visually richer and more diverse samples.

6.5. Impact of synthetic image generation for balancing the dataset with stable diffusion

By using Stable Diffusion-generated images, it is observed that dust detection on solar panels has significantly improved by addressing the imbalance in the dataset. Stable Diffusion enhanced the performance of the models in dust detection under various conditions. The models were able to learn more diverse and complex patterns by looking at these generated images, which are accurate representations of the types of dirt and faults that would be found in real-world solar panel conditions. Consequently, these models trained on the balanced dataset had a better generalization ability. Another factor was that the synthetic images were realistic, which helped the model recognize dirty panels more accurately, especially when such images were less frequently represented. As a result, higher precision, recall, and F1-score were achieved for the dirty panels trained with models on imbalanced datasets shown in Figs. 13, 15, and 16. From the results, it is evident that dust detection on solar panels is significantly improved by applying Stable Diffusion on the imbalanced dataset. Finally, due to the high accuracy and balanced prediction in both dirty and clean panel cases within our study, Stable Diffusion performed best among the evaluated methods for handling imbalance, suggesting it is a strong candidate for PV-dust detection.

6.6. Effectiveness of data augmentation to address imbalanced datasets

The effectiveness of Stable Diffusion and SMOTE in addressing the data imbalance issues is clearly demonstrated in our results and observations. SMOTE balanced the actual dataset by generating minority class samples, allowing the model to predict dust accumulation on solar panels more accurately. At the same time, Stable Diffusion generated high-quality real-time images to balance the dataset. Stable Diffusion helped increase accuracy, precision, and recall for both clean and dirty panels, according to the model, which enabled better generalization. In the imbalanced dataset scenario, as expected, models had a strong bias for predicting the clean panels (Hong et al., 2020). The problem was, however, mitigated after applying SMOTE and Stable Diffusion. Specifically, ResNet50 was the best model, and the ones that achieved the highest accuracy and performed the best in detecting dirty panels were the models based on Stable Diffusion. In contrast, VGG-16 had the best performance on both SMOTE and Stable Diffusion. These improved F1-scores and recall for dirty panels indicate that both SMOTE and Stable Diffusion helped the models learn more balanced features more effectively. The observations from these results confirm that the approach effectively addresses the data imbalance problem by utilizing Stable Diffusion and SMOTE to enhance the prediction of dust accumulation on solar panels.

6.7. Impact of cross-validation on model reliability

Compared to our initial single train-test split results, the cross-validation metrics are consistently 3–4 % lower for most models and metrics, as expected for a more conservative evaluation. This reduction demonstrates that our updated protocol avoids overfitting and provides a realistic estimate of generalization. Notably, all models retain the same trends as in the original results, with substantial improvements observed when using SMOTE and Stable Diffusion to address class imbalance, as discussed in Table 10. The low standard deviations further indicate stable performance across all folds, confirming that our models do not overfit to any particular split or the minority class. Among the tested models, VGG-16 and ResNet50 achieved the highest overall performance after balancing, especially when using Stable Diffusion. These trends are consistent with prior studies, which indicate that careful data augmentation and rigorous evaluation protocols are essential for reliable model deployment in real-world solar panel monitoring scenarios (Haque et al., 2025).

6.8. Real-world deployment with imbalanced dataset

The robustness evaluation conducted in 5.3 provides further evidence that the proposed augmentation methods, particularly Stable Diffusion, remain effective under real-world conditions where data distributions are naturally imbalanced. While augmentation is often validated only on balanced splits, our experiments confirmed that models trained on balanced data retained strong performance when tested directly on the original imbalanced sets. In particular, Stable Diffusion consistently improved both Accuracy and F1-score across architectures, with ResNet50 showing the largest improvements in accuracy and F1-score. These findings demonstrate that the proposed framework is not only effective in controlled evaluations but also resilient to deployment conditions, reinforcing its practical utility for PV monitoring applications.

6.9. Reflection on significance testing and robustness

The inclusion of statistical significance testing provides important validation that the improvements observed with Stable Diffusion are not attributable to random fold-to-fold variation, which is a particular concern when working with modest datasets. The Wilcoxon Signed Rank analysis (see Table 11) confirmed that Stable Diffusion achieved statistically significant gains over both Imbalance and SMOTE for VGG-16 and ResNet50, while MobileNetV3 demonstrated smaller but consistent improvements. These results highlight two key points. First, the superiority of Stable Diffusion holds across architectures, with especially strong benefits for deeper networks such as ResNet50, where both Accuracy and F1-score gains were highly significant ($p < 0.01$). Second, the variability in MobileNetV3 outcomes illustrates that augmentation benefits may be model-dependent and influenced by backbone capacity. Nevertheless, we acknowledge that the relatively small dataset size increases variance across folds, and while statistical testing mitigates this limitation, it does not eliminate it entirely. Future work will therefore focus on validating these findings using larger, real-world PV datasets to further establish the generalizability of the proposed augmentation framework.

6.10. Evaluation using κ and MCC

The consistent improvement in κ and MCC, as shown in Table 12, across different balancing strategies indicates the consistent performance and generalizability of the classification framework. While traditional metrics, such as accuracy and F1-score, indicate model improvements, the inclusion of κ and MCC—both of which account for chance agreement and class imbalance—demonstrates that these gains represent genuine advances in predictive performance. Notably, the near-perfect κ and MCC values observed with Stable Diffusion indicate that the models are capable of making highly reliable and balanced predictions across both classes, which is especially critical for real-world solar panel dust detection, where accurate detection of the minority class is essential.

6.11. Computational complexity and cost effectiveness

The computational complexity and cost-effectiveness of the proposed solar panel fault detection workflow were systematically evaluated across both data augmentation and model inference stages. Using the Stable Diffusion XL pipeline for synthetic image generation on an NVIDIA RTX 4060 GPU, 162 images were produced in approximately 63 min, averaging 23.39 s per image with peak GPU and RAM usage of 6.57 GB and 6.8 GB, respectively. Each image generation follows a complexity of $O(T \cdot F)$, where T represents the number of diffusion steps and F the computational cost per neural network forward pass, confirming that high-quality augmentation is feasible and resource-efficient on consumer hardware. For model complexity, the VGG-16, ResNet50,

and MobileNetV3 architectures comprised 14.8 M, 24.2 M, and 3.3 M parameters, respectively. Training times per epoch ranged from 11.20 to 14.54 s, and inference times ranged from 0.58 to 1.24 s, validating their computational tractability. Collectively, the integration of advanced image-level augmentation with optimized CNN models enables improved detection accuracy without excessive hardware or time costs, demonstrating a scalable and operationally efficient solution suitable for practical solar panel maintenance and monitoring scenarios.

6.12. Comparison with prior studies

In Table 13, a comparison of the proposed models with the best previous techniques that utilize different augmentation strategies is presented. However, prior studies have highlighted the shortcomings of standard oversampling approaches, particularly for image-based tasks, despite the use of generative models like Stable Diffusion, which has achieved accuracies of 96 % (Sousa et al., 2024) and 95.40 % (Sagers et al.,) in previous studies, as shown in Table 6. Earlier methods relied on SMOTE or GANs; however, the integration of Stable Diffusion resulted in highly realistic synthetic images that preserve the spatial features necessary for CNN performance. Our proposed models, specifically ResNet50, achieved 98.87 % accuracy, whereas VGG-16 achieved 93.50 % accuracy with SMOTE augmentation. Meanwhile, with Stable Diffusion, VGG-16 performed well, achieving an accuracy of 97.64 %. This helped achieve significant model robustness and generalization. The results demonstrate that the effective synergistic combination of the advanced generative augmentation with the CNN model achieves unprecedented dust detection accuracy in solar panel maintenance and renewable energy optimization.

6.13. Limitations of SMOTE on image data

SMOTE is commonly used to address class imbalance; however, it was originally intended for tabular data. In this study, SMOTE is applied by reshaping image data into a vector form to facilitate comparative analysis. Although this process discards important spatial relationships between pixels, the loss of spatial integrity may compromise the process's ability to perform well in other tasks, such as image classification. Hence, it is intended to use SMOTE as a baseline benchmark for our experiments since it is not an optimal augmentation method for image-based CNN models. Future work attempts to improve oversampling techniques such as DeepSMOTE, which will utilize feature embeddings from the deep CNN layers to preserve spatial structure in the sampled data. In this work, SMOTE was therefore included not as a competing augmentation strategy but as a classical, widely cited baseline to provide a point of comparison with image-level generative methods. This design choice allows the study to demonstrate why spatially consistent augmentation, such as Stable Diffusion, is necessary for image tasks. Including SMOTE highlights the relative advantage of generative augmentation, while ensuring fairness by benchmarking against a

standard oversampling technique that remains prevalent in the literature on imbalance.

6.14. Assumptions, limitations, and uncertainties

This section explicitly states the key assumptions underlying our methodology, acknowledges study limitations, and identifies sources of uncertainty that may affect interpretation and generalizability of the results.

6.14.1. Assumptions

- Stable Diffusion-generated images approximate real-world dust accumulation patterns sufficiently for effective classifier training.
- SMOTE-based feature interpolation in vector space provides a reasonable approximation of the minority class distribution, despite loss of spatial information.
- Class labels (clean/dusty) in the SolNET dataset are accurate and consistently assigned.
- Pre-trained CNN weights (ImageNet) transfer effectively to the PV panel inspection domain.
- The two-stage validation protocol (visual screening + FID/KID/pHash) adequately ensures synthetic image quality.

6.14.2. Limitations

- Single dataset: Experiments rely solely on the publicly available SolNET dataset from Bangladesh.
- PV technology: Only polycrystalline silicon (poly-Si) modules are represented; monocrystalline, thin-film, and bifacial technologies are excluded.
- Binary classification: The study addresses only clean vs. dusty classification, not multi-class fault detection.
- Unknown acquisition metadata: Camera specifications, capture distance, irradiance, temperature, and time of capture are not documented in the dataset.
- Uniform resizing: All images resized to 224×224 pixels, which may reduce fine-grained micro-texture information relevant to dust detection.
- Modest dataset size: 842 images (502 clean, 340 dusty) limits training diversity despite cross-validation mitigation.

6.14.3. Uncertainties / Threats to validity

- Domain gap: Potential mismatch between synthetic image characteristics and real-world soiling physics may affect model generalization.
- Cross-validation variance: Modest dataset size introduces fold-to-fold variability in performance estimates.

Table 13
Comparison table of proposed research results with prior studies.

Ref	Dataset	CNN Models	Augmentation Technique	Accuracy
(Onim et al., 2022)	Solar Panel Dataset	VGG-16, SolNet, ResNet50, InceptionV3, AlexNet	SMOTE	90.2 % on SolNet
(Alatwi et al., 2024)	Solar Panel Dataset	Multiple CNN versions of VGG, ResNet, MobileNet, etc.	GAN-based	86.79 % (Max on DenseNet169)
(Özdemir et al., 2021)	Xuzhou HYSPEX	CNN Models	SMOTE and ADASYN	94.46 %
(Mehta and Saini, 2024)	Solar Panel Dataset	CNN, GAN	VGG-16 (89.2), InceptionV3 (90.5), CNN-GAN (96.2)	
(Sagers et al.,)	Skin Disease Image Data	CNN and ViT (Vision Transformer)	Diffusion Model	95.4 % with CNN
(Sousa et al., 2024)	Earth Observation Image Data	ResNet50 and ViT-B/32	Stable Diffusion	66 % and 96 %
Proposed Approach	Solar Panel Dataset	ResNet50	Stable Diffusion	98.87 %
Proposed Approach	Solar Panel Dataset	VGG-16	SMOTE	93.50 %

- Prompt sensitivity: Stable Diffusion outputs depend on prompt design; alternative prompts may yield different augmentation quality.
- Filtering subjectivity: Manual screening of synthetic images involves subjective judgment, potentially introducing bias.
- Environmental variability: Real-world deployment conditions (seasonal changes, varying illumination, different dust compositions) are not fully represented.

These factors should be considered when interpreting results and planning future validation studies on larger, multi-technology PV datasets. PV module appearance and geometry can vary substantially across technologies and manufacturers (cell layout, busbar configuration, and framing), which leads to different visual patterns in inspection imagery. This technology-driven variability has been highlighted in regional PV technology mapping studies, which show that different PV technologies are deployed across locations and climates, motivating robustness to heterogeneous module types in practical monitoring systems (Nassar et al., 2022). In this study, this variability was controlled by restricting experiments to the SolNET dataset, which contains only polycrystalline silicon (poly-Si) modules. Therefore, the proposed results primarily demonstrate the effectiveness of imbalance mitigation within a consistent module-technology domain. Generalization to heterogeneous PV module types and viewpoints remains an open deployment factor and will be addressed in future work through multi-technology training data, domain adaptation, and augmentation strategies that explicitly model viewpoint and geometric diversity.

7. Conclusion and future directions

7.1. Conclusion

In this research, the issue of data imbalance in solar panel dust detection is addressed by applying two advanced data augmentation techniques: SMOTE and Stable Diffusion. The study aimed to enhance the performance of convolutional neural networks (CNNs) for detecting dust on solar panels and improving the maintenance and efficiency of solar energy systems. With SMOTE and Stable Diffusion, the dataset is balanced by generating synthetic samples of the minority class, thus mitigating the model's bias towards the majority samples (clean class). The results showed that SMOTE and Stable Diffusion improved the models' performances. However, the results of Stable Diffusion surpassed the performance of models in detecting panels with dirt. The results illustrated in Fig. 11 show that the models of VGG-16, ResNet50, and MobileNetV3 performed significantly better using the SMOTE and Stable Diffusion technique; specifically, VGG-16 performed very well by achieving 93.5 % accuracy. In the case of Stable Diffusion, ResNet50 outperformed VGG-16, achieving 98.87 % accuracy in predicting dust on solar panels, whereas VGG-16 achieved 97.64 % accuracy. These models generalized better across synthetic images from Stable Diffusion, which was able to produce more realistic synthetic images and handle the minority class. Furthermore, this research highlights the need for utilizing augmentation techniques such as Stable Diffusion in real-world applications involving imbalanced datasets. This proposed research aims to reduce overall costs by enhancing solar panel maintenance and energy efficiency through more accurate dust detection. In real-world testing, where the training set was balanced with SMOTE or Stable Diffusion and the test set remained imbalanced, the proposed models still achieved robust performance, demonstrating strong generalization ability. For instance, VGG-16 achieved 98 % accuracy, ResNet50 reached 96 %, and MobileNetV3 obtained 96 % on the realistic test set as discussed in Fig. 19. The adoption of stratified 10-fold cross-validation significantly enhanced the robustness and credibility of our experimental findings, addressing the risks of overfitting and optimistic bias associated with small and imbalanced datasets. The consistency and low variance of our results across all folds confirm the effectiveness and

reliability of our proposed data-balancing strategies. Future work will further strengthen the generalizability of our approach by incorporating additional external validation datasets, conducting ablation studies on augmentation parameters, and making the code and synthetic data publicly available to support open science and reproducibility. The application of Cohen's kappa and MCC provided a rigorous and balanced validation of the performance of our models. The high values achieved, particularly under Stable Diffusion balancing, confirm that the proposed approach not only improves overall accuracy but also delivers reliable, class-balanced predictions. These findings underscore the practical reliability of our method for deployment in real-world solar panel monitoring systems. These results confirm the practical effectiveness of advanced augmentation techniques in addressing class imbalance and supporting reliable, automated solar panel maintenance in operational environments. Integrating advanced data augmentation techniques, such as Stable Diffusion, creates new possibilities for making an automated solar panel maintenance system. Our approach can help automate cleaning schedules, optimize energy output, and reduce maintenance costs in solar farms by improving the model's ability to detect dust automatically. The main assumptions and uncertainties (dataset scope, module-type restriction, synthetic data fidelity, and evaluation variance) are discussed in 6.14. This research has practical applications in developing low-cost, real-time monitoring systems that produce actionable insights, which can be integrated into existing solar panel networks to enhance performance.

7.2. Future directions

In future research, we plan to further enhance the robustness and generalizability of our approach by investigating advanced hybrid augmentation strategies that integrate oversampling (SMOTE and Stable Diffusion) with undersampling techniques such as Tomek Link. This hybrid framework will be systematically evaluated to determine its effectiveness in mitigating class overlap and noise in highly imbalanced datasets. Furthermore, we intend to explore the application of explainable AI methods such as Grad-CAM and SHAP to provide deeper interpretability of the CNN models' predictions, thereby facilitating transparent and trustworthy deployment in operational solar panel monitoring systems. Future work will extend the framework to multi-technology PV datasets (mono-Si, thin-film, bifacial) with diverse module geometries and viewpoints to validate robustness under heterogeneous visual domains.

Dataset availability

The dataset employed in this study is publicly accessible online, with the source cited in the bibliography (Onim et al., 2022).

Code availability

The code for data preprocessing, model training, evaluation, and implementation for this study will be released on a public GitHub repository (Kumar, 2025) upon acceptance.

Ethical approval

This study did not involve human participants or animal experiments; hence, no ethical approval was required.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors acknowledge the use of Grammarly for grammar checking and sentence refinement during the preparation of this manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Notation and Abbreviations

To aid readability, Table A1 summarizes the abbreviations and notations used throughout this paper.

Table A1	
List of Abbreviations and Notations	
Acronym	Expanded Form
AI	Artificial Intelligence
CIA	Controllable Image Augmentation
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CNNs	Convolutional Neural Networks
CV	Cross-Validation
DDIM	Denoising Diffusion Implicit Model
DDPM	Denoising Diffusion Probabilistic Model
DDPMs	Denoising Diffusion Probabilistic Models
DeepSMOTE	Deep Synthetic Minority Over-Sampling Technique
DR	Diabetic Retinopathy
GAN	Generative Adversarial Network
GANs	Generative Adversarial Networks
LoRA	Low-Rank Adaptation
MCC	Matthews Correlation Coefficient
ML	Machine Learning
MNIST	Modified National Institute of Standards and Technology (database)
MSIT	Ministry of Science and ICT
NAS	Neural Architecture Search
NL	Nonlinear
NRF	National Research Foundation
PV	Photovoltaic
RB	Rough Set Theory
ReLU	Rectified Linear Unit
ResNet	Residual Network
ResNet50	Residual Network 50
SD	Stable Diffusion
SDXL	Stable Diffusion XL
SE	Squeeze-and-Excitation
SMOTE	Synthetic Minority Over-Sampling Technique
SMOTE-RSB	Synthetic Minority Oversampling Technique based on Rough Set Theory
SVM	Support Vector Machine
SolNet	Solar Network
VGG	Visual Geometry Group
VGG-16	Visual Geometry Group 16
VIT	Vision Transformer
FID	Fréchet Inception Distance
KID	Kernel Inception Distance
pHash	Perceptual Hashing
Notations	
Symbol	Description / Usage
%	Percentage (e.g., accuracy values like 97.6 %)
±	Plus-minus (e.g., mean ± std. dev., like 95.6 % ± 1.7 %)
°	Degree (e.g., rotations like ±40°)
×	Multiplication (e.g., image dimensions like 224 × 224)
*	Multiplication or corresponding author indicator (e.g., xnew = xi + δ(xi − xi))
−	Minus or dash (e.g., in ranges like 1–34, or formulas like po − pe)
/	Division or slash (e.g., TP / (TP + FP))
=	Equals (e.g., Accuracy = (TP + TN) / (TP + TN + FP + FN))
√	Square root (e.g., in MCC formula √(((TP + FP)(TP + FN)...))
^	Hat (e.g., xi for selected neighbor)
δ	Delta (e.g., random factor between 0 and 1 in interpolation)
κ	Kappa (e.g., Cohen’s kappa symbol)
→	Arrow (e.g., Data Collection → Imbalanced Dataset)

Data availability

I have shared the link of data/code in my manuscript.

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